Building Segregation: The Long-Run Neighborhood Effects of American Public Housing *

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Abstract

Public housing construction caused large, persistent increases in Black population shares and substantial declines in median incomes and rents in American neighborhoods. This paper studies these long-term neighborhood effects of the American public housing program using a new national dataset tracking over 1 million public housing units built between 1935 and 1973, linked to neighborhood-level data from 1930 to 2010. I document that projects were systematically targeted towards poorer neighborhoods with higher Black population shares, reflecting the program's racialized site selection. Using a stacked matched difference-in-differences approach, I compare treated neighborhoods to matched controls within the same county. Geographic spillovers to nearby neighborhoods were modest, with small increases in Black population shares driven by white population decline. Effects were concentrated in neighborhoods with initial Black shares in a plausible tipping range, where public housing triggered substantial white outflows. Finally, children from low-income families who grew up in public housing neighborhoods experienced significantly lower rates of upward mobility. These findings demonstrate that mid-century public housing, despite intentions of slum clearance, reinforced existing patterns of racial and economic segregation with lasting consequences for economic opportunity.

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1 Introduction

> The Congress hereby declares that the general welfare and security of the Nation and the health and living standards of its people require housing production and community development sufficient to remedy the housing shortage, eliminate substandard housing through the clearance of slums and blighted areas, and achieve as soon as feasible the goal of a decent home and suitable living environment for every American family.
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> — *Housing Act of 1949*

Between the 1930s and 1970s, the United States constructed approximately 1.4 million federally-funded public housing units in one of the most ambitious urban policy initiatives of the 20th century (Schwartz 2021). Designed to clear slums, address urban housing shortages, and provide affordable housing to low-income families, the American public housing program has, to many, become synonymous with policy failure. Prominent accounts in urban history and sociology have argued that the program created and entrenched racial and economic segregation (e.g., R. Rothstein 2017; Hirsch 1998; Massey and Denton 1993), and accelerated mid-century urban decline (Jackson 1985). Large, megablock projects such as Cabrini-Green, Pruitt-Igoe, and the Robert Taylor Homes became infamous, criticized for concentrating poverty, promoting crime, and destroying the urban fabric of neighborhoods (Jacobs 1961; Newman 1972). Federal policy since the 1970s has reflected this negative view through a retreat from public housing: The government has shifted funding toward market-based alternatives like Housing Choice Vouchers and the Low Income Housing Tax Credit, while simultaneously dismantling the existing stock through the HOPE VI program, which has funded the demolition or removal of roughly 30% of the original federal public housing stock since the early 1990s (Schwartz 2021).

Yet the evidence for public housing's failure is mixed. Some scholars contend that the program has been unfairly maligned by its most notorious examples and argue its poor reputation partially reflects broader social and economic challenges faced by American cities in the postwar period (Bauman 1994; Bloom, Umbach, and Vale 2015; Goetz 2013). Indeed, despite widespread criticism of the program, approximately two-thirds of public housing residents reported being "satisfied" or "very satisfied" with their housing as late as 1999 (Schwartz 2021). Some causal evidence from economists suggests that living in public housing either has neutral or positive effects on individual outcomes (Chaudhry and Eng 2024; Currie and Yelowitz 2000; Jacob 2004; Pollakowski et al. 2022), which could indicate that public housing may not have been as detrimental as some narratives suggest. At the same time, disentangling the local effects of public housing from pre-existing neighborhood conditions remains challenging. Given that projects were often sited in already-distressed areas as part of slum clearance efforts, did public housing cause neighborhood decline, or simply reflect the targeting of struggling areas? A complete understanding of the public housing program has high stakes as cities grapple with housing affordability: The question of what role public housing can and should play in addressing urban housing challenges remains highly relevant.

Resolving these debates requires systematic evidence on where public housing was built and how it shaped neighborhoods over the long run. Yet such an analysis has been hindered by the lack of a comprehensive public dataset linking projects to their construction dates and precise locations. This paper addresses this research gap by constructing a new dataset on American public housing. Combining previously digitized data, public data sources, and newly digitized materials, I obtain the locations, construction dates, and characteristics, such as project size and racial composition, of over 8,000 public housing projects containing over 1 million units nationwide. I then integrate this dataset with a consistent-area panel of Census tract-level data from 1930-2010 that I build using both existing Census tract data and geolocated full count Census data from 1930 and 1940, all concorded to consistent tract boundaries. I also merge these data with three other sources: Home Owners' Loan Corporation (HOLC) "redlining" maps from the late 1930s, maps with the locations of Urban Renewal projects, and the locations of interstate highways. This new dataset enables me to examine several fundamental questions about the siting and long-run neighborhood consequences of the U.S. public housing program.

First, what neighborhood characteristics influenced public housing site selection? This is important for understanding the political economy of public housing and for informing my identification strategy for estimating the neighborhood effects of public housing construction. In particular, I am interested in whether public housing was explicitly targeted towards Black neighborhoods, as some historical accounts suggest (e.g., Hirsch 1998; R. Rothstein 2017), or whether it was more broadly targeted towards poor and working-class neighborhoods. To answer this, I estimate linear probability models to explore how pre-existing neighborhood characteristics, including demographic, socioeconomic, housing, and institutional factors, predict the placement of public housing projects. I find that public housing projects were more likely to be built in neighborhoods that were initially poorer, more populated, and had higher shares of Black residents, consistent with the program's slum clearance goals and the racialized politics of housing policy in the mid-20th century. I also find that neighborhoods that received public housing were more likely to be affected by urban renewal, another large mid-century urban policy that targeted predominantly Black neighborhoods (LaVoice 2024). Finally, I find that public housing projects built in poorer neighborhoods and neighborhoods with higher

Black population shares tended to have higher shares of Black residents themselves, indicating that public housing reinforced rather than disrupted existing residential segregation patterns. I also show that these patterns hold in a specific instance of site selection in Philadelphia by digitizing a map of proposed-but-not-selected public housing sites from Bauman (1987): In particular, neighborhoods proposed but rejected for public housing were initially much whiter than those that ultimately received projects. These results are consistent with historical narratives and case study evidence about the politics of public housing site selection (e.g., Meyerson and Banfield 1955; Bauman 1994; Hirsch 1998).

Second, what were the short- and long-run effects of public housing construction on both the recipient and surrounding neighborhoods? To answer this question, I employ a stacked matched difference-in-differences strategy that explicitly compares each public housing neighborhood to a matched never-treated control neighborhood based on pre-treatment characteristics. For each neighborhood that received a public housing project, I use nearest-neighbor propensity score matching to identify a comparable control neighborhood in the same county based on the pre-treatment characteristics that I demonstrated were key determinants of site selection. I then treat each matched pair as a separate "sub-experiment" and stack these sub-experiments into a single analytic dataset. The specification includes matched-pair-by-year fixed effects, ensuring that each treated neighborhood is compared only to its matched control in each year, while controlling for time-varying shocks that affect both neighborhoods in the same pair. Identification stems from the fact that federal funding for public housing was limited, and local housing authorities could not build projects in every eligible neighborhood, creating variation in project placement across otherwise similar areas. Importantly, this stacked framework also avoids the econometric pitfalls that arise in staggered adoption settings when treatment effects are heterogeneous across cohorts and over time (Wing, Freedman, and Hollingsworth 2024). I extend this methodology to analyze the geographic spillovers of the projects by conducting a similar exercise for neighborhoods adjacent to public housing neighborhoods, thereby estimating the broader geographic effects of the projects.

In the public housing ("treated") neighborhoods, I find large increases in total population driven particularly by increases in the Black population, resulting in substantial increases in Black population shares. I also find large decreases in median rents, along with declines in median incomes and various other measures of economic well-being, including lower labor force participation rates and higher unemployment rates. These results, in conjunction with the site-selection results, suggest that while public housing was targeted toward poorer, minority neighborhoods, the construction of the projects further accelerated demographic changes and socioeconomic decline in these neighborhoods over the long run.

I find modest geographic spillovers to adjacent neighborhoods: median incomes declined and Black population shares increased by 2-4 percentage points, driven primarily by white population decline rather than Black population inflows.

I also explore the heterogeneity of these effects along several dimensions. First, I explore heterogeneity based on the neighborhood's initial racial composition. I find that treated and nearby neighborhoods that initially had Black population shares within a potential "tipping range" (between 1% and 12%, based on work by Card, Mas, and J. Rothstein 2008) saw outflows of white residents. In contrast, neighborhoods that were initially more Black did not. This suggests that public housing construction triggered racial tipping dynamics in some neighborhoods. I also find that the long-run effects of public housing construction on neighborhoods were larger for projects built before 1960, and that the impact of the projects I detect was driven mainly by neighborhoods not treated by the urban renewal program.

Finally, I explore the implications of these neighborhood effects for economic opportunity. To do so, I link my dataset to data on the upward mobility of children born to low-income families born from 1978-1983 from Opportunity Atlas (Chetty et al. 2018). I find that low-income children growing up in neighborhoods that received public housing projects experienced significantly lower upward mobility than children in matched control neighborhoods, even after controlling for neighborhood characteristics. I also find small adverse effects on upward mobility in nearby neighborhoods, but these effects are fully accounted for by controlling for neighborhood income and demographics in 1980, with no additional effects from proximity to public housing itself. This pattern suggests that these modest spillovers occurred through earlier neighborhood changes, rather than through persistent or independent effects of public housing on later mobility outcomes in surrounding neighborhoods.

This paper contributes to several literatures in urban and public economics and economic history. First, it contributes to the literature on the local effects of affordable housing programs (Baum-Snow and Marion 2009; Diamond and McQuade 2019) and revitalization policies (Collins and Shester 2013; LaVoice 2024; Rossi-Hansberg, Sarte, and Owens 2010) by estimating the neighborhood effects of one of the largest urban policies in American history. More directly, recent literature has studied the effects of HOPE VI public housing demolitions on neighborhood (Tach and Emory 2017; Blanco and Neri 2025; Aliprantis and Hartley 2015; Sandler 2017; Almagro, Chyn, and Stuart 2023) and individual (Haltiwanger et al. 2024; Chyn 2018) outcomes. This literature has focused on the demolitions of the most distressed public housing projects, and much of it has been focused on a single city (Chicago). There are several reasons to think these estimates of the effects of demolitions may not generalize to those of public housing construction. First, the projects demolished through the HOPE VI program are a selected set of the most distressed projects. They are thus not necessarily representative of the public housing program as a whole. Second, the effects of project construction likely depended on existing neighborhood conditions, which likely varied in ways that they did not for demolitions. New projects may have served as local amenities in some neighborhoods but disamenities in others. Finally, the effects in Chicago might not necessarily generalize to other cities.

My findings also complement earlier work from other social sciences on public housing and the concentration of poverty in central cities, which generally used data compiled from one or a small number of cities (e.g. Carter, Schill, and Wachter 1998; Massey and Kanaiaupuni 1993). Relative to this older literature, I use a dataset covering a much broader set of cities, projects, and time periods, and apply modern econometric techniques to estimate the causal effects of public housing construction on neighborhoods.

My paper adds to the small literature estimating the effects of public housing construction. Shester (2013) uses digitized data from HUD that contain all federally funded public housing projects built until 1973, with locations at only the locality level (discussed in Section 4) to study the effect of public housing construction on a variety of county- and city-level outcomes. This paper finds that cities in the same state with more public housing construction experienced decreases in median property values, median family income, and population density. Shester, Allen, and Handy (2019) uses the same dataset to study the effects of public housing construction on the rise in single motherhood at the MSA level. I build on this work by merging precise locations to these data, allowing me to study neighborhood effects.

The most closely related contribution is that of Guennewig-Moenert (2025), who studies the effects of public housing construction in New York City, specifically on neighborhood racial composition and rents, along with welfare estimates based on a structural model. I compile data on public housing projects across the country, enabling me to characterize the effects of public housing in the United States more broadly and to speak to broader historical debates about the program. Scholars have argued that the New York City Housing Authority (NYCHA) was in many ways exceptional in its effectiveness at dealing with many of the factors that have plagued public housing authorities in the U.S. (Bloom 2008).[1] My empirical approach also differs: he uses variation in proximity to public housing to define the control group, while I use similarity in pre-treatment characteristics. Another related contribution is contemporane-

[1]For example, NYCHA was a relatively small participant in HOPE VI and did not demolish any of its high-rise projects through the program (Schwartz 2021).

ous work by Harris (2025), who studies the short-run neighborhood effects in the first two decades of the program and finds that public housing initially reduced poverty concentration in recipient neighborhoods.

Finally, my paper contributes to the economic literature on the emergence and consequences of segregation in the United States. While prior research has documented migration responses to neighborhood demographic change (Shertzer and Walsh 2019; Boustan 2010), my paper addresses one way in which federal housing policy directly influenced neighborhood sorting. More broadly, this paper contributes to the literature and debates around what role public policy has played in shaping racial and economic segregation in American cities (R. Rothstein 2017; Trounstine 2018; Boustan 2013; T. D. Logan and Parman 2025).

The paper proceeds as follows. Section 2 describes the history of the public housing program. Section 3 outlines the potential channels through which public housing may affect neighborhoods. Section 4 describes data sources and construction. Section 5 analyzes the determinants of public housing site selection. Section 6 presents the empirical strategy and estimates of neighborhood effects. Section 7 explores heterogeneity in these effects and examines potential mechanisms. Section 8 links public housing to long-run economic opportunity using data from the Opportunity Atlas. Section 9 concludes.

2 Historical Background: The U.S. Public Housing Program

Public housing in the United States began during the Great Depression as part of New Deal efforts to reduce urban housing shortages and stimulate the economy. The Public Works Administration built the first projects, producing about 20,000 units before court challenges over land acquisition limited its scope (Jackson 1985).

The program was greatly expanded and decentralized with the passage of the Housing Act of 1937, which promoted the creation of local Public Housing Authorities (PHAs) to build and oversee public housing with federal funds. Specifically, federal grants covered the difference between operating costs and tenant revenue, while PHAs were responsible for site selection and project management. The 1937 Act's stated purpose was slum clearance and redevelopment rather than expanding housing supply, which reflected contemporary beliefs that "slum" neighborhoods reduced nearby property values and caused poor health and behavior among residents (Meyerson and Banfield 1955). Approximately 170,000 housing units were constructed under this act, with nearly 90% built on former slum sites (Schwartz 2021).

The program was greatly expanded as part of a broad urban policy agenda following World

War II. Many Northern cities experienced severe housing shortages, which were especially acute for African Americans, while white flight, suburbanization, and urban "blight" challenged cities. These challenges generated a large political coalition for urban redevelopment policy, culminating in the Housing Act of 1949 (von Hoffman 2000). Title III of the 1949 Act provided funding for an additional 810,000 units of public housing and marked a major shift in the program's scope and goals, expanding public housing as a tool for slum clearance, a source of low-income housing, and a potential destination for those displaced by the Urban Renewal slum clearance program, which was funded by Title I of the Act.

The ambitious goals of the 1949 Act, however, quickly met political, fiscal, and social obstacles. Several developments in the 1950s and 1960s transformed public housing into a deeply controversial policy.

First, as detailed in Section 2.1, racial dynamics in site selection caused intense political conflicts. Second, the composition of public housing tenants changed significantly over time. Early projects primarily housed working families and the "deserving poor," but by the 1960s, they increasingly housed the poorest and most disadvantaged households. The rise of suburban homeownership allowed working-class families to leave public housing, while authorities tightened income eligibility limits. Federal regulations also mandated that PHAs prioritize the neediest applicants, resulting in the concentration of poverty within projects (Schwartz 2021). Third, design and construction issues became more evident. To avoid competing with private housing and to meet mandated construction cost limits, projects were intentionally made austere and inexpensive, making them susceptible to rapid deterioration (Schwartz 2021). Many public housing authorities struggled to fund maintenance as middle-class tenants exited the projects. Furthermore, the 1969 Brooke Amendment capped tenant rents at 25% of income, further limiting PHA revenues (Hunt 2018). These challenges are illustrated by well-known public housing failures, such as St. Louis's Pruitt-Igoe Homes, built in 1954. Initially praised as a model development, Pruitt-Igoe quickly faced issues like crime, vandalism, and abandonment as maintenance funding dried up and middle-class residents moved out (Bristol 1991). As a result of these issues, the complex was ultimately demolished beginning in 1972.

In response to mounting criticism, American housing policy has shifted away from direct public housing provision. A 1971 report by the Nixon Administration wrote that "drab, monolithic housing projects, largely segregated...still stand in our cities as prisons of the poor" (Orlebeke 2000). In 1973, President Nixon declared a moratorium on subsidies for traditional public housing. By the early 1990s, the public housing stock faced serious challenges: physical deterioration due to deferred maintenance, extreme concentration of poverty as working families moved out, and, in some cases, rampant crime and gang activity. The federal response to these challenges was the HOPE VI program, launched in 1992, which provided federal funding to demolish distressed projects or transform them into mixed-income developments. The Faircloth Amendment of 1998 further limited public housing growth by capping the total number of units PHAs could operate at their 1999 levels.

2.1 Race, Segregation, and Public Housing

Race and segregation emerged as central and contentious issues throughout the program's history. From the outset, the Public Works Administration followed a so-called "neighborhood composition rule," formally segregating projects based on the demographics of the neighborhoods where they were built. While ostensibly a neutral policy intended to maintain neighborhood "stability", in practice the rule solidified segregation. Notable accounts, such as R. Rothstein (2017), contend that public housing not only reinforced existing segregation but also actively created it by constructing segregated projects in previously integrated neighborhoods. A well-known example is Atlanta's Techwood Homes, built in 1936 as an all-white development, which displaced a previously integrated low-income neighborhood (R. Rothstein 2017). Many large public housing authorities continued to follow an explicit neighborhood-composition rule well into the 1960s.[2] However, even after these legal changes, projects remained highly segregated in practice (Bickford and Massey 1991).

Historical case studies show that issues related to race and segregation also influenced site selection decisions. Additionally, contemporaneous debates among policymakers reveal that they were fully aware of the program's potential to either worsen or improve residential segregation patterns (Hirsch 2000). In Chicago, the Housing Authority initially planned to build projects throughout the city, but strong opposition from white neighborhoods led to the concentration of projects in predominantly Black areas on the South and West Sides (Meyerson and Banfield 1955; Hirsch 1998). Similar conflicts occurred in other major cities, with white residents and politicians successfully blocking projects in their neighborhoods. I examine a specific example of this dynamic in Philadelphia, comparing proposed-but-not-selected areas to public housing locations in Section 5.2. Building on this historical evidence, Section 5.1 tests whether these racial site selection dynamics held systematically nationwide.

[2] Although legal challenges began in the mid-1950s, including the U.S. Supreme Court's 1954 decision to let stand a California ruling striking down San Francisco's segregated projects, federal law did not formally prohibit race-based segregation in public housing until the Fair Housing Act of 1968.

2.2 Motivating Example: Hunters Point in San Francisco

The history of Bayview–Hunters Point in San Francisco provides a clear example of how public housing construction can change a neighborhood's long-run trajectory. Before World War II, the area was a racially mixed neighborhood on the city's periphery. During the war, the federal government built thousands of temporary housing units there to accommodate defense workers, causing an initial influx of African American families. After the war, instead of dismantling these temporary units, local and federal officials converted many of them into permanent public housing. Throughout the 1950s and 1960s, the San Francisco Housing Authority replaced and expanded these units into large-scale projects and directed Black families into these developments. This expansion coincided with a quick demographic shift: as the projects grew, the neighborhood ceased to be a mixed-income area; white and Asian households left, and the community became increasingly Black and segregated. Figure 1 illustrates the racial transition of these neighborhoods along with the location of public housing projects, showing the sharp rise in the Black population share and the decline in the white population from 1940 to 1970. The rest of this paper tests whether the patterns seen in Hunters Point were systematic across the country.

3 Potential Effects of Public Housing

In this section, I outline the main channels through which the construction of public housing projects might affect neighborhood outcomes.

First, public housing construction may mechanically alter the racial and socioeconomic composition of neighborhoods. Tenant selection policies, income limits, and subsidized rents determined who occupied these projects, thereby altering the demographic makeup of the areas where they were built.

Second, these compositional changes could trigger endogenous sorting by private households. If individuals have preferences for their neighbors' race or income, the arrival of public housing residents might prompt some existing residents to relocate. Historical work, such as Jackson (1985), argues that part of mid-century "white flight" from central cities was a response to the public housing program. However, empirical evidence on this point remains limited.

Third, the physical structure of public housing projects could create built-environment externalities. Replacing substandard or vacant structures with new housing could improve neighborhood quality and increase neighborhood desirability, as shown for private housing upgrades by Rossi-Hansberg, Sarte, and Owens (2010) and for subsidized housing by Ellen et al. (2007). Conversely, large superblocks or architecturally incongruous high-rise towers might be viewed as local disamenities, and indeed, criticism of the mid-century public housing program often focused on the design of the projects (Jacobs 1961; Newman 1997).

Fourth, public housing construction may generate broader social and market externalities. A frequent concern is that concentrated poverty within large projects generated social disorder and elevated crime, with potential spillovers into surrounding neighborhoods (Sandler 2017). Conversely, visible public investment could signal neighborhood revitalization and spur private reinvestment, improving local conditions (Rossi-Hansberg, Sarte, and Owens 2010). The direction of these effects may vary across contexts. Modern evidence on the LIHTC program suggests that subsidized housing investments may be a positive amenity in distressed areas but may be perceived negatively in more affluent ones (Diamond and McQuade 2019).

I explore these mechanisms empirically in Section 7.

4 Data

4.1 Public Housing Data

A significant challenge in studying the history of public housing is the lack of a comprehensive dataset that includes project lists, construction dates, and precise project locations. Consequently, previous research on the neighborhood effects of public housing has been limited in scope or restricted to a small number of cities where researchers could obtain this information directly from housing authorities. The data limitations have also hindered broader historical analyses of the public housing program, and have been acknowledged by previous work (Ellen et al. 2007; Hunt 2018). My paper addresses this gap by constructing what I believe to be the most complete dataset of mid-century public housing by combining information from a series of administrative datasets, along with previously digitized and newly digitized sources.

The first source I use is the *Consolidated Development Directory* (CDD), published by the US Department of Housing and Urban Development (HUD) in 1973 and digitized by Shester (2013). This data contains the universe of federally funded public housing projects that existed in 1973, along with various project characteristics (e.g., number of units) and, crucially, the year each project was completed. However, the CDD does not contain location information for the projects. Previous work using these data (Shester 2013; Shester, Allen, and Handy 2019) has therefore been limited to studying aggregate city- and county-level outcomes. These data also contain three project number fields, which I combine with the state code to form federal project codes in the standard HUD format (e.g., IL-2-38-1 for a Chicago project). These project codes allow me to link the CDD data to other project-level datasets such as the Picture of Subsidized Households and HUD Form 951. As far as I know, this element of these data has not been previously exploited.

To obtain location information for the projects, I turn to a series of publicly available HUD administrative datasets. The primary of these is the Picture of Subsidized Households (PSH) datasets, which contain a list of federally funded housing projects, various project characteristics (e.g., number of units, demographics), and, from 1997 onward, their locations. I use the PSH datasets from 2000 and 1997, due to limitations in each. I also link to more recent National Data Geospatial Asset (2023) data. Finally, I link to the HUD File 951 dataset, which lists street addresses, latitudes, and longitudes for the stock of multifamily assisted housing projects that existed between 1986 and 1995 (Kucheva 2013). These datasets largely overlap in coverage, but each contains projects missing from the others. This linkage allows me to assign locations to over 90% of the projects in the CDD. I will refer to this as the geocoded CDD dataset.

I supplemented these data with hand-collected information from historical annual reports of local public housing agencies, obtained from various libraries (or directly from the housing authority in the case of San Francisco), and from FOIA requests to public housing agencies. From this effort, I was able to obtain supplementary data from eight major cities: New York, Chicago, Boston, Los Angeles, Washington, DC, San Francisco, Atlanta, and Baltimore.[3] For these cities, I collected the complete set of projects built up to 1973, including their construction dates and locations. I was able to geolocate these projects using the Google Maps API. For projects that I was unable to geolocate successfully, I hand-identified locations based on street address and name.

There were two motivations for collecting these additional data. First, it allowed for a better understanding of the projects in the CDD for which I was unable to assign locations. Public housing demolitions, such as those from the HOPE VI program that began in 1993, may have caused some CDD projects to be absent from later administrative datasets used for geocoding, preventing me from assigning locations to those projects in the merged dataset. Indeed, the incomplete matching between CDD and PSH data suggests that some projects may have ceased to exist when PSH data collection began. Moreover, these missing projects might not be randomly distributed, as demolitions targeted particularly blighted projects.

Second, the CDD-HUD data include only federally funded public housing projects. Little

[3]I collected digitized reports from other cities, such as Cleveland, Cincinnati, and St. Louis, but not all contained the precise project locations information.

data exist on city- and state-funded public housing projects, but some cities may have funded some public housing construction outside of the federal program. New York City, in particular, has a notable city- and state-funded public housing program, and using only the data on federal projects in New York City misses a substantial number of housing projects and units.[4] The other cities for which I collected data had either none or very few city or state-funded projects. Ultimately, I use my hand-collected data for housing projects in New York City, Chicago, and San Francisco, supplement the geocoded CDD for Boston and Washington, DC, and rely on the geocoded CDD data for all other cities (Los Angeles, Atlanta, and Baltimore).[5] I found little evidence in the historical record suggesting that, by otherwise relying on data on federal projects, I am missing a notable stock of public housing in other cities.

The last step in constructing the public housing dataset is to obtain information on the populations and racial composition of the projects. These data allow me to distinguish between public housing residents and the rest of the neighborhood's population. For most cities, I obtain this information from the 1977 Picture of Subsidized Households dataset, which was cleaned and shared with me by Yana Kucheva (Kucheva 2013). These data report the number of subsidized households by race in each project, but do not include direct population counts. In Appendix A, I describe how I convert these household counts to population counts. I match these data to the geocoded CDD dataset using federal project codes. For New York City, I used population-by-race data from the early 1970s digitized and shared with me by Max Guennewig-Moenert (Guennewig-Moenert 2025). And for Chicago, I obtained population-by-race data from the 1973 digitized Annual Report of the Chicago Housing Authority. While it would be ideal to measure population by race at the time of construction, these data are not available in most cities. Thus, I assume the population and racial composition of each project in each decade after construction is equal to the 1970s values. I use these population estimates to estimate the private (non-public-housing) population in each decade by subtracting the estimated public-housing population from the total Census-reported population.

The result of this process is a dataset containing the construction dates, coordinates, and characteristics of over 8,000 projects and 1 million units of public housing built from 1935 until 1973.

[4] My digitized dataset of NYC housing projects contains 129,430 housing units, compared to 80,000 in the PSH-CDD data.

[5] For Los Angeles, Atlanta, and Baltimore, I find no additional projects in my digitized data versus the geocoded CDD data.

4.2 Neighborhood Data

To study the neighborhood effects of public housing construction, I construct a panel dataset of neighborhood-level characteristics spanning 1930 to 2010. Building such a dataset presents several challenges. First, the number of cities with tract-level data is limited in 1940 and especially in 1930. Second, Census tract boundaries change over time, requiring the construction of a consistent panel. And third, income data was not collected in the 1930 Census, and median income was not reported in the tract-level Census tables in 1940.

I proceed as follows. First, I collect a variety of outcomes at the Census tract level from the 1930 to 2010 decennial censuses. These data include tract-level population counts by race (white, Black, and other), several socioeconomic measures (median income, high school graduation rates, labor force participation, and unemployment rates), and median rents and home values. All census data and shapefiles were acquired from IPUMS NHGIS (Manson et al. 2022).

I construct a consistent panel of census tracts by concording all tracts to 1950 Census tract boundaries using an area-reweighting approach. I describe the tract harmonization in more detail in Appendix B.2. I chose 1950 as the base year for two reasons. First, to limit concerns about results being driven by public housing-driven changes in tract boundaries, I chose a year early in my analysis period, rather than at the end, as much of the literature does. Second, publicly available 1940 New York City census tract shapefiles do not correspond to actual census tracts, but to much larger health districts, making 1940 a less suitable base year. Given that New York City is a major city in my sample, I chose 1950 as the base year for the entire sample.

I supplement this tract-level data with data from the 1930 and 1940 full-count Census (Ruggles et al. 2022). To convert the full count data to the tract level, I first aggregate the individual-level data to the enumeration district (ED) level using the 1930 and 1940 enumeration district shapefiles from the Urban Transitions Project (J. R. Logan et al. 2024). I then use area reweighting to aggregate these data to 1950 Census tracts. Incorporating these full-count data serves two purposes. First, they allow me to expand my set of cities and neighborhoods for which tract-level data was unavailable in 1930 and 1940. Second, they allow me to include income information for 1930 and 1940, which is not available in the tract-level tables for those years. I proxy for income in 1940 and 1930 using full-count Census microdata. For 1940, I sum individual wage income (INCWAGE) across all household members to calculate total household wage income, then classify households into income bins within each enumeration district. For 1930, I use machine-learning-adjusted occupation scores from Saavedra and Twinam (2020) aggregated to the household level and similarly binned. In both years, I use income bins that match the 1950 census income categories. I concord these enumeration district-level data to 1950 Census tracts using area-reweighting and calculate tract-level medians from the binned distributions. The area-reweighting and median calculation procedures are described in Appendix B.2.

I convert all monetary values to 2000-dollar values using the US CPI from Officer and Williamson (2025). For population counts, monetary variables, and distance measures, I apply the inverse hyperbolic sine (asinh) transformation in much of the analysis. The asinh transformation approximates the natural logarithm for large values while being defined at zero. This transformation compresses the potentially skewed distributions of these variables and allows regression coefficients to be interpreted approximately as percentage changes without requiring me to drop tracts with zero population in some racial groups (Bellemare and Wichman 2020).[6] When reporting percentage changes from asinh-transformed outcomes in the text, I convert regression coefficients using the hyperbolic sine function, $\sinh(\beta) \times 100$.

In addition to Census data, I incorporate several historical datasets that capture other urban policies that may be related to public housing placement and subsequent neighborhood change. First, I incorporate digitized data and shapefiles of the Home Owners' Loan Corporation (HOLC) "redlining" maps drawn in the late 1930s, made available by *Mapping Inequality* (Nelson and Winling 2023). These maps graded neighborhoods from "A" (best) to "D" (hazardous) to indicate perceived mortgage risk. Although many scholars have argued that these maps institutionalized redlining and curtailed investment in minority neighborhoods (Aaronson, Hartley, and Mazumder 2021), recent evidence complicates this view. Fishback et al. (2024) shows that the maps largely reflected existing patterns of racial and socioeconomic segregation rather than shaping new lending decisions. For my purposes, these maps serve as a historical snapshot of neighborhood conditions and perceived credit risk in the late 1930s and thus serve as a proxy for pre-existing discrimination, rather than a direct driver of exclusion. I overlay these maps onto the 1950 Census tract boundaries. Following Weiwu (2025), I classify a neighborhood as "redlined" if at least 80% of the area is designated as "hazardous" (HOLC grade D).

Second, I also incorporate data on the locations of U.S. urban renewal projects (1955-1966) from *Renewing Inequality* (Nelson and Ayers 2025). Urban renewal, established under Title I of the 1949 Housing Act, provided federal subsidies for cities to acquire and clear "blighted" neighborhoods. Urban renewal was closely intertwined with public housing: clearance projects often created sites for new developments or displaced families that public housing was intended

[6] I demonstrate that results are qualitatively similar when using outcome variables in levels in Section 6.4.

to rehouse (von Hoffman 2000). Recent research finds that Black neighborhoods were two to three times more likely to receive urban renewal projects than white neighborhoods, and that urban renewal areas experienced declines in housing and population density alongside rising rents and incomes (LaVoice 2024). To account for potentially confounding effects of urban renewal and to explore the relationship between the two programs, I overlay these maps onto the 1950 Census tract boundaries and classify a tract as an "urban renewal" tract if more than 5% of its area overlaps with an urban renewal project.

Finally, I incorporate data on the locations of interstate highways as of 1996 from Weiwu (2025). Interstate highway construction began in the late 1950s, and has been accused of disproportionately displacing minority communities (Rose and Mohl 2012). These data allow me to control for potential confounding effects of highway construction on neighborhood change, as well as to explore the relationship between highway locations and public housing placement.

4.3 Defining Treatment and Spillover Neighborhoods

To study the neighborhood effects of public housing construction, I define a set of *treated* tracts that received public housing projects and a set of *nearby* tracts that may have experienced spillover effects from nearby projects. To ensure that a project represents a meaningful neighborhood intervention, I include only projects with at least 50 housing units.

Because the public housing projects are geocoded at the coordinate level, I use a geographic buffer approach to define treated tracts. Specifically, a tract is defined as treated if any portion of it intersects a 100-meter buffer around a public housing project. This definition captures cases in which a project straddles multiple tracts; when that occurs, I allocate its units and population evenly across the affected tracts.

Since the Census data are decennial, I assign treatment timing by decade. Following the literature, I use the project's completion date as the treatment date (Asquith, Mast, and Reed 2023). A tract is considered treated in year t if its first qualifying project was completed between $t-9$ and t (e.g., tracts receiving projects between 1951 and 1960 are treated in 1960). This timing convention ensures that the treatment occurs after the pre-treatment observation but before the post-treatment observation in the panel. [7] For tracts receiving multiple public housing projects over time, treatment timing is determined by the first project meeting the size thresholds. Subsequent projects in the same tract are not considered separate treatment

[7] If a project was completed in the Census year but after the Census enumeration date, the effects of the project would meaningfully be captured in the second, rather than first, post-treatment decade, and the $t = 0$ effects would be understated.

events, since the tract has already been "treated" by public housing construction.

To examine spatial spillovers, I define a nearby tract as one that shares a border with a treated tract, is not itself treated, and lies within one kilometer of the nearest public housing project. This hybrid contiguity-and-distance definition identifies neighborhoods that are close enough to plausibly experience externalities from nearby developments while excluding tracts that are technically adjacent to project neighborhoods but geographically distant from a project. I adopt a one-kilometer threshold based on prior literature showing the geographic extent of spatial spillovers from public housing interventions (Blanco and Neri 2025). Nearby tracts inherit the earliest treatment year among their treated neighbors. Figure 2 illustrates the treated and nearby classification for Chicago, showing treated tracts and their adjacent spillover areas.

4.4 Sample Selection

My empirical analysis focuses on projects built between 1941 and 1973, with neighborhood outcomes measured from 1930 to 2010. I restrict the sample in several ways to ensure a balanced panel of neighborhoods with sufficient pre-treatment data to conduct my difference-in-differences analysis.

Table 1 shows the impact of each filtering step on the sample composition. Starting from the original sample of 12,062 census tracts representing 38.1 million people in 1940 (approximately 29% of the U.S. population), I apply the following restrictions. First, I require that tracts exist in all years from 1930 to 2010, which drops about 25% of tracts. Second, I drop tracts for which I cannot calculate population by race, median income, median rent, labor force participation rates, and unemployment rates for all years, which removes an additional 5% of tracts. Third, I exclude tracts with public housing built before 1941 to preserve 1930 and 1940 as clean pre-treatment periods, which removes 109 treated tracts containing 109 projects. Fourth, I drop metropolitan areas with fewer than 30 total tracts to ensure sufficient within-city variation. This excludes three small metropolitan areas. Finally, I exclude population outliers: tracts with populations below the 5th percentile or above the 98th percentile in any year.

The resulting balanced sample includes 6,507 census tracts across 47 CBSAs, representing 27.5 million people in 1940 (approximately 21% of the U.S. population). This sample contains 811 public housing projects with 299,900 total units located in 820 treated tracts. Figure 3 shows the geographic distribution of CBSAs included in the analysis. These sample restrictions have implications for external validity: Due to data limitations, the analysis focuses on neighborhoods in major metropolitan areas that had established census tracts by 1930, excluding public housing that was built in rural areas, smaller cities, or neighborhoods that were not populated in the beginning of the period. By excluding projects with fewer than 50 units, my analysis also does not speak to the effects of smaller-scale or scattered-site public housing.

5 Site Selection

In this section, I study the placement of the public housing projects in my sample. Understanding these dynamics is essential for several reasons. First, the site-selection process for mid-century public housing was a significant criticism of the program. Historical case studies in several cities suggest that projects were often targeted to poorer and minority neighborhoods, both because of the program's slum clearance goals and because of local backlash in white neighborhoods against construction of projects (Meyerson and Banfield 1955; Bauman 1994; Sugrue 2005). Second, these site selection dynamics have had important legal implications: Lawsuits in several cities have alleged that public housing site selection was discriminatory and in violation of Civil Rights Law, most famously Gautreaux v. Chicago Housing Authority (1966), but also in cities like Dallas (Walker v. HUD 1985) and Baltimore (Thompson v. HUD 2005). Still, there has been little systematic evidence on the nationwide patterns of public housing site selection. Finally, understanding these site-selection dynamics is important for estimating the neighborhood effects of public housing and interpreting those effects.

To begin, in Section 5.1, I estimate which pre-existing neighborhood characteristics predict the eventual locations of public housing projects. I also test whether the placement of public housing projects was related to the locations of other mid-century urban policies, particularly urban renewal and the interstate highway system. Then, in Section 5.2, I zoom in on a particular case study of site selection by examining the locations of proposed-but-not-built public housing in Philadelphia in Section 5.2. This example illustrates the political dynamics around site selection decisions.

Finally, in Section 5.3, I examine whether the racial composition of the projects themselves varied systematically with the initial characteristics of the neighborhoods in which they were built.

5.1 Where were the projects built?

Table 2 shows the raw comparison of 1940 baseline characteristics between census tracts that eventually received a public housing project and those that did not. Treated neighborhoods had substantially higher Black population shares, lower median incomes and rents, higher unemployment rates, a higher share of homes needing major repairs, and were located closer to a central business district.

To formally test which neighborhood characteristics predict public housing placement, I estimate linear probability models relating pre-treatment tract characteristics to the likelihood of ever receiving a public housing project. Specifically, I estimate:

$$\text{Treated}_{ic} = \gamma_c + \boldsymbol{X}'_{i,1940}\boldsymbol{\beta} + \boldsymbol{Z}'_i\boldsymbol{\delta} + \epsilon_{ic} \tag{1}$$

where Treated_{ic} is an indicator for whether tract i in county c received any public housing project between 1941 and 1973. The vector $\boldsymbol{X}_{i,1940}$ contains 1940 neighborhood characteristics, and $\boldsymbol{Z}_i$ includes distance to the nearest interstate highway and an indicator for whether a tract was treated by urban renewal. I include county fixed effects γ_c to absorb systematic differences in placement across local public housing authorities.

Columns 1-3 of Table 3 estimate the model without the $\boldsymbol{Z}_i$ controls, while Column 4 adds them to examine whether the relationship between 1940 characteristics and public housing placement is robust to controlling for subsequent federal interventions. I adjust standard errors for spatial correlation following Conley (1999), allowing for correlation of residuals across census tracts within a 2-kilometer radius.

I chose variables based on historical narratives of public housing site selection and the program's stated intentions. First, as described in Section 2, the public housing program was largely intended as a "slum clearance" program, so we might expect that neighborhoods with lower socioeconomic status and worse housing market conditions would be more likely to receive public housing. Second, historical case studies in multiple cities have documented the key role of race in determining where public housing was built (e.g. Hirsch 1998; Bauman 1994), so I include the Black population share as a key predictor of public housing placement. I also include the HOLC redlining designation to capture long-term patterns of racial segregation, discrimination, and disinvestment.

The results from these linear probability models are shown in Columns 1-4 in Table 3. Column 1 shows a parsimonious model with several key neighborhood characteristics, while Column 2 shows a more saturated model including all neighborhood characteristics. Column 3 adds the share of housing units deemed in need of major repairs in 1940, which is missing for about 5% of tracts. Consistent with the historical narrative, I find that public housing projects tended to be targeted towards poorer, minority neighborhoods: Census tracts that were initially more populated, had higher Black population shares, lower median incomes and labor force participation rates, and had higher unemployment rates were more likely to receive public housing projects during this period. Public housing was also more likely to be built in neighborhoods that redlined, had lower rents, and had a higher share of homes needing repairs, although these estimates do not reach statistical significance in the fully saturated models. These findings reflect the slum clearance motivation of the public housing program and are also consistent with historical accounts emphasizing the racial targeting of public housing siting: Even after controlling for local economic and housing market conditions, the Black population share remains a strong and significant predictor of public housing placement.

Column 4 additionally tests whether neighborhoods that ultimately received public housing were more likely to be affected by two other transformative mid-century urban policies: urban renewal and the interstate highway system. Evidence and historical narrative suggest that these programs may have affected many of the neighborhoods that were also targeted for public housing. The Urban Renewal program, in particular, was directly related to public housing in many cities, as public housing projects were used to house individuals displaced by urban renewal (Bauman 1994; Hirsch 1998). Neighborhoods that received public housing were also much more likely to be affected by urban renewal, whereas proximity to an interstate highway was not significantly related to public housing placement. This result further confirms the interplay between these two programs and motivates accounting for urban renewal in my empirical strategy in Section 6.[8]

Quantitatively, these effects are sizable. The baseline probability that a census tract in the sample received a public housing project between 1941 and 1973 is 12.6%. Based on the estimates in Column (4), a one standard deviation increase in Black population share increased public housing selection probability by 3.3 percentage points (26.4% increase), a one standard deviation increase in unemployment rate increased the probability by 4.4 percentage points (34.7% increase), and a one standard deviation decrease in median income increased the probability by 2.1 percentage points (16.5% increase). Finally, neighborhoods that were eventually targeted by urban renewal were 8.8 percentage points (70%) more likely to receive public housing.

5.2 A Philadelphia Case Study

Historical accounts of public housing site selection have highlighted the political battles that surrounded the placement of projects in particular cities, for example, in Chicago (Meyerson and Banfield 1955), Philadelphia (Bauman 1994), and Detroit (Sugrue 2005). These ac-

[8] These results are consistent with similar exercises in Harris (2025) and Massey and Kanaiaupuni (1993).

counts have emphasized the conflict between public housing authorities and white working-class neighborhoods that resisted the construction of public housing projects in their communities. Here, I present a case study of site selection in Philadelphia, drawing on the historical treatment and maps from Bauman (1987). This case study is illustrative of the political dynamics that shaped site selection decisions in many cities (Hunt 2005).

Following the 1954 Housing Act, the Philadelphia Housing Authority proposed 21 public housing projects across the city. Upon announcement, many of these proposed projects faced significant backlash from local white communities, leading to their eventual cancellation. I identified these proposed Philadelphia public housing locations by scanning and georeferencing a historical map from Bauman (1987), shown in Figure 4. I georeferenced the historical map in QGIS, aligning it with a modern basemap using identifiable landmarks, such as major roads and rivers. Then, I located the proposed public housing sites on the georeferenced map and matched these points to 1950 census tracts. In total, I identify 12 neighborhoods with proposed-but-not-built public housing sites in Philadelphia.

I then compare the baseline characteristics of proposed-but-not-built project locations to those of actual public housing locations in Philadelphia. Table 4 presents the balance table, which shows that the initial characteristics of the proposed locations were very different from those of actual public housing locations. Most notably, these proposed-but-not-built project locations had very low initial Black population shares. They were also further from the central business district, less populated, and had lower unemployment rates. These differences illustrate the dynamics of site selection in one particular city, where whiter neighborhoods often resisted public housing construction. They also show that an identification strategy based on proposed-but-not-built locations would likely be invalid, as these locations likely do not represent a good counterfactual for actual public housing locations.

5.3 Did project demographics vary with neighborhood characteristics?

I now examine how the racial composition of the public housing projects varied with the initial neighborhood characteristics in which they were built. The key question is whether projects reinforced existing patterns of residential segregation or disrupted them. I focus on the share of Black residents in each project, using data from treated census tracts in my balanced sample for which project-level racial composition is available from the 1970s. Ideally, I would observe the racial composition of the projects at the time of construction, but I do not have this information. Still, to the extent that the racial composition of the projects was relatively stable over time, the 1970s data should provide a reasonable proxy for the initial demographics of the projects.

Formally, I regress the project Black share in each tract i in county c on the neighborhood characteristics measured in the decade before construction:

$$\text{Project Black Share}_{ic} = \gamma_c + \boldsymbol{X}'_{i,t-1}\boldsymbol{\beta} + \epsilon_{ic} \tag{2}$$

The unit of analysis is the tract, with one observation per treated tract. For the 75% of treated tracts containing a single public housing project, Project Black Share$_{ic}$ is simply the Black share in that project. For tracts with multiple projects, I define t as the construction decade of the first project to open in the tract, and Project Black Share$_{ic}$ is the population-weighted average Black share across all projects that opened in that first decade. Whereas the regressions in Section 5.1 used 1940 covariates (or later urban policy exposure) for all projects, here I match each tract to neighborhood characteristics from the decade immediately preceding its construction to better capture local conditions at the time of development. The results are shown in Table 5. Column (1) presents a parsimonious specification with only baseline Black share and median income, column (2) includes a fuller set of neighborhood characteristics, including rent, population, unemployment, distance to the CBD, and redlining status, while column (3) adds county fixed effects. I again include the distance from an interstate highway and urban renewal designation.

The core result is robust across specifications: public housing projects largely matched the racial composition of their surrounding neighborhoods, reinforcing rather than disrupting existing residential segregation patterns. Based on the coefficient in column (3), a 10 percentage point increase in baseline neighborhood Black share predicts a 2.7 percentage point increase in project Black share. Furthermore, projects built in poorer neighborhoods, even conditional on race, tended to be more heavily Black, as indicated by the negative coefficient on median income and positive coefficient on unemployment rate.

6 The Effect of Public Housing Construction on Neighborhoods

Having established in the previous section that public housing projects were systematically targeted towards poorer, minority neighborhoods, I now turn to estimating the effects of the projects on neighborhood change in subsequent decades.

6.1 Research Design

I employ a stacked matched difference-in-differences approach to estimate the causal effects of public housing on neighborhood outcomes in both the treated and nearby neighborhoods. The approach matches each treated or nearby tract to an observationally similar never-treated neighborhood, then compares changes within these matched pairs over time while avoiding contaminated comparisons across different treatment cohorts and controlling for time-varying shocks that affect both neighborhoods in each matched pair. This design addresses two key challenges: selection bias from non-random project placement and econometric issues from staggered treatment timing. The implementation proceeds in four steps:

Step 1: Defining the Donor Pool. I restrict potential controls to census tracts that never received public housing and are never nearby tracts, as defined in Section 4. This allows me to separately estimate direct effects and geographic spillovers, while ensuring clean controls and reducing concerns about SUTVA violations.

Step 2: Tract-Specific Propensity Score Matching. For each treatment year cohort, I match both directly treated and nearby tracts to controls from the donor pool. For example, for the 1950 treatment cohort, both the directly treated tracts and their nearby tracts are matched based on their 1930 and 1940 characteristics. This ensures control neighborhoods followed similar pre-treatment trajectories over the relevant decades. Within each cohort, I implement nearest-neighbor matching with replacement using propensity scores estimated on the key predictors of site selection from Section 5, measured in each of the two pre-treatment decades: total population, Black population share, median income, unemployment rate, and labor force participation rate. Each treated or nearby tract is matched to its closest control based on propensity score distance. I also require exact matches on urban renewal designation and county. Exact matching on urban renewal ensures that the control neighborhoods were subject to the same urban renewal policies as the treated neighborhoods, thereby avoiding confounding the effects of the two policies. Exact matching on county also ensures that treated and control tracts are subject to the same local economic shocks and policy environment. The within-county restriction leverages the fact that federal funding for public housing was limited, and that public housing authorities therefore did not build in every eligible neighborhood within their jurisdiction, creating plausibly exogenous variation in treatment conditional on observables. These restrictions mean that 16 of the 820 treated tracts in the balanced panel could not be matched to a suitable control within their county and urban renewal status, leaving 804 treated tracts in the matched sample.

Figure 5 shows absolute standard mean differences across pre-treatment characteristics for the treated neighborhoods and their matched controls. The matching achieves reasonably good balance on all pre-treatment characteristics, though given the restrictiveness of exact matching on county and urban renewal status, some covariates have standardized differences above 0.1. Section 6.4 shows that results are robust to alternative matching procedures that achieve better balance, whether by dropping poor matches or matching across metropolitan areas.

Step 3: Construction of Stacked Matched-Pair Dataset. Following Wing, Freedman, and Hollingsworth (2024), I create a stacked analytic dataset where each matched pair forms an independent "sub-experiment" observed from 1930 to 2010. Within each sub-experiment, the treated (or nearby) tract and its matched control are assigned a common treatment year, the year the treated neighborhood received public housing, and unique matched-pair identifiers. Control neighborhoods may appear in multiple sub-experiments if they serve as matches for multiple treated neighborhoods.

Step 4: Stacked Event-Study Estimation. I stack all matched-pair sub-experiments and implement a stacked difference-in-differences design that compares changes in outcomes over time between each treated neighborhood and its matched control. In particular, for tract i in matched pair m in time t, I estimate the following event-study specification:

$$y_{imt} = \alpha_{im} + \sum_{\tau \neq -1} \beta_\tau (D^\tau_{imt} \times \text{Treated}_i) + \delta_{mt} + \varepsilon_{imt} \tag{3}$$

where y_{imt} is the outcome of interest for tract i in matched pair m at time t, α_{im} are tract-by-pair fixed effects, $D^\tau_{imt} = \mathbb{1}[(t - T_{im}) = \tau]$ are indicators for event time τ relative to treatment year, Treated_i indicates whether tract i is a treated tract, and δ_{mt} are matched pair-by-year fixed effects. Since all fixed effects are implemented within a sub-experiment defined by each matched pair, identification solely relies on within-matched-pair comparisons. This specification implements the stacked difference-in-differences framework of Wing, Freedman, and Hollingsworth (2024), where I use matched pairs to define the sub-experiments. While Wing, Freedman, and Hollingsworth (2024) demonstrate the benefits of stacking to avoid negative weights in staggered adoption settings, my approach combines their framework with propensity score matching to address selection and staggered timing challenges simultaneously.[9]

The matched-pair-by-year fixed effects (δ_{mt}) provide particularly fine-grained controls for potential confounders. Because matching creates pairs of observationally similar neighborhoods within the same county, these fixed effects absorb not only county-level shocks but also shocks that differentially affect specific types of neighborhoods. For instance, if deindustri-

[9] Recent work by Blanco and Neri (2025) and Guennewig-Moenert (2025) use stacked difference-in-differences designs to study public housing interventions, using only distance to select the control group.

alization particularly impacted working-class, Black neighborhoods in Chicago in 1980, and both tracts in a matched pair are working-class, Black Chicago neighborhoods, this shock is absorbed by the pair-by-year fixed effect.

The tract-by-pair fixed effects (α_{im}) absorb all time-invariant differences within each matched pair, including unobserved characteristics that may have influenced both site selection and outcomes. Together, this fixed effects structure ensures identification comes exclusively from differential changes within pairs of similar neighborhoods.

The β_τ coefficients capture the average treatment effect at event time τ across all matched pairs. For directly treated tracts, this identifies the effect of hosting public housing. For nearby tracts, it identifies geographic spillover effects. Because each treated unit is matched to a single control unit, I weight each observation equally, yielding a simple average across sub-experiments without the weighting adjustments outlined in Wing, Freedman, and Hollingsworth (2024). Standard errors are adjusted for spatial correlation following Conley (1999) within a radius of 2 kilometers, allowing for dependence between nearby tracts. In Appendix C, I show that results are robust to using tract-level clustered standard errors instead.

The identifying parallel trends assumption operates at the matched-pair level: within each pair, the treated (or nearby) tract would have continued evolving like its matched control absent public housing. While this assumption is fundamentally untestable, three pieces of evidence support its plausibility. First, the event study specification allows me to examine pre-treatment trends over two decades, and I find no systematic differential trends between treated and control neighborhoods within matched pairs prior to public housing construction. Second, while Section 5.1 demonstrates that neighborhood demographic and economic conditions influenced site selection, these observable characteristics only explain about 11% of the variation in placement decisions. This suggests substantial idiosyncratic variation in which specific neighborhoods received projects among observationally similar candidates. Third, Section 6.4 demonstrates that results are robust to alternative matching procedures and specifications, including stricter matching criteria and cross-metro comparisons that relax the common local shocks assumption while allowing for closer matches. Together, this evidence suggests that within matched pairs of similar neighborhoods, the selection of which neighborhood received public housing provides plausibly exogenous variation for identifying causal effects.

6.2 Effects on Treated Neighborhoods

Figure 6 presents the effects of public housing construction on the inverse hyperbolic sine of population by race. Following public housing construction, the total population in the public housing neighborhoods increased substantially, rising by approximately 12.5%[10] in the immediate decade following construction, increasing to 14-15% in the following decades. This increase was driven by substantial increases in the total Black population, a 27.7% increase relative to the matched control three decades after construction. On average, we see little to no effect on the white population. These significant population increases reflect the influx of public housing residents themselves: Figure 7 shows results for the non-public housing population (the difference between the total and public housing populations) and shows a large average decline in the private population, both white and Black, following public housing construction.

Figure 8 shows that these population changes led to changes in the racial composition of the treated neighborhoods. Relative to the control neighborhoods, Black population shares increased by 2.6 percentage points in the immediate decade following construction, and continued to increase up to approximately 5.0 percentage points three decades after construction. The latter estimate represents an 18.0% increase in the Black population share relative to the baseline share of 27.8%.

Figure 9 shows the effects of public housing on median rent and income, showing broad declines in both. Median incomes fall sharply following public housing construction and decline further in the long run: median income falls by 7.7% immediately after construction, declining to 15.3% three decades after construction. Median rents decline over time, with statistically insignificant declines immediately after construction but declining by 14.5% three decades after construction.

6.3 Effects on nearby neighborhoods

A primary source of backlash against the mid-century public housing program has been the concern that it precipitated a broader urban decline and white flight, with negative spillovers on surrounding communities (Jackson 1985). Understanding these potential spillovers is crucial to assessing the program's overall urban impact. To test this, I estimate the geographic spillover effects of public housing on nearby neighborhoods.

Using the matched nearby tracts defined in Section 6.1, I estimate geographic spillover effects of public housing on surrounding neighborhoods. Balance statistics for the nearby tract matches (Figure 10) show good balance on all pre-treatment characteristics, with only Black population showing a standardized mean difference of slightly above 0.1.

[10]The event study figures display raw asinh coefficients. Percentage changes reported in the text are calculated as $\sinh(\beta) \times 100$, which approximates $\beta \times 100$ for small coefficients and provides a more accurate conversion for larger effects (Bellemare and Wichman 2020).

I then estimate the treatment effects for these nearby neighborhoods using the same stacked difference-in-differences design as in Equation 3.

Figure 11 shows the effects of public housing construction on population by race in the nearby neighborhoods. In contrast to the treated neighborhoods, I find that the spillover neighborhoods experienced small population declines overall. I find statistically significant declines in white population in the long run, but minimal changes in Black population. Consequently, while Black population shares increased by approximately 2-4 percentage points in nearby neighborhoods, this change appears to be driven primarily by white population decline rather than Black population growth, in contrast to the treated neighborhoods where substantial Black population inflows occurred.

Figure 12 shows the effects of public housing construction on median rent and income in the nearby neighborhoods. I find small but statistically significant declines in median income in nearby neighborhoods, with median income falling by 3.8% in the first decade following construction and by 2.3% in the second decade. Effects in later years become statistically insignificant, though the point estimates remain negative. This may indicate some degree of economic decline or sorting in these nearby neighborhoods. However, I find no effect on median rent in these nearby neighborhoods, suggesting that the housing market effects of public housing construction were not widespread.

This evidence suggests that public housing construction did affect nearby neighborhoods by precipitating white flight. However, the modest effects on median incomes and lack of effects on rents suggest that the economic dimensions of "urban decline" attributed to public housing in historical narratives (Jackson 1985) were more limited than the demographic changes.

6.4 Robustness Checks

One might worry that the results are sensitive to the specific matching specification. In particular, one might be concerned that the exact matching criteria are too restrictive, resulting in imperfect nearest-neighbor matches. Indeed, the balance achieved in the main specifications above is imperfect, with some standardized mean differences between 0.1 and 0.2. I run several alternative matching specifications to test the robustness of these results. First, I run a specification that drops poor matches: in particular, I apply a caliper that excludes matches with a propensity score difference of more than 0.2 standard deviations. Second, I loosen the exact-match restriction on county and instead require a match from a neighborhood within any other metropolitan area. This expands the donor pool, particularly for treated neighborhoods in smaller cities. A comparison of the results from these alternative specifications is shown in Figure 13 for treated neighborhoods and Figure 14 for nearby neighborhoods. Overall, the estimates are largely similar across these alternative specifications, both in sign and magnitude.

Another potential concern is that the use of the inverse hyperbolic sine transformation for population and monetary outcomes could affect the interpretation of results. Recent work by Chen and Roth (2024) highlights that logarithmic transformations in difference-in-differences designs can produce estimates that are sensitive to the choice of units and may conflate extensive and intensive margin effects. McConnell (2024) further shows that when treated and control groups have different baseline means, log specifications can yield misleading estimates and even produce opposite signs relative to levels specifications. To address this concern, I re-estimate all specifications using outcome variables in levels rather than asinh-transformed values. The results show that both specifications yield qualitatively identical conclusions: all outcomes show the same sign of treatment effect. Furthermore, when the level coefficients are expressed as a percentage of the baseline control mean—an approach suggested by Chen and Roth (2024) to ensure unit-invariance—the magnitudes are broadly consistent with the asinh estimates. For example, median income declined by 15.1% in the levels specification compared to 15.3% in the asinh specification, while Black population increased by 26.2% (levels) versus 37.5% (asinh). These results confirm that my findings are not sensitive to the choice of functional form.

One more fundamental concern with this design is that, if individuals are re-sorted across neighborhoods as a result of the construction of public housing, then the within-metro control neighborhoods may in fact be affected by the treatment as well. For example, one might be concerned that, in response to the construction of public housing, individuals who initially lived in a treated neighborhood might relocate to the initially similar control neighborhood. This would be a violation of the stable unit treatment value assumption (SUTVA).

I show two pieces of evidence that these potential SUTVA violations are not driving my results. First, I run a specification in which control neighborhoods are only selected from metropolitan areas with low public housing intensity. In these cities, control areas are less likely to be affected by any potential spillovers from public housing. I calculate each CBSA's intensity as total public housing units divided by the total population of the neighborhoods in that metropolitan area in my balanced sample in 1950. Across the 47 metropolitan areas in my sample, the intensity ranges from 0 to 20 units per 1,000 people, with a median of 7.0. I restrict the donor pool to metros in the bottom quintile of intensity, which includes large metros such as Los Angeles, Detroit, St. Louis, and Indianapolis. I then match treated and nearby neighborhoods from all metros to controls in these low-intensity metros, requiring that matches come from different metropolitan areas.

If SUTVA violations were inflating my baseline estimates—for example, because Black residents leaving control tracts made those controls appear less Black, thereby exaggerating the treatment-control differences—then I would expect smaller effects using these less-contaminated controls. Instead, I find that the effects on Black population shares are larger than my baseline. These results are presented in Appendix Table 17.

Second, if results are driven by SUTVA violations, we might expect offsetting movements between the control tracts and treated tracts. For example, we might see the Black population share in matched controls decline as residents move to the treated neighborhoods, which could drive the event study results. Appendix Figure 22 plots raw Black shares for treated neighborhoods and their matched controls over event time. Black shares increase in both treated and control neighborhoods, but more so in the treated neighborhoods. The absence of declining Black shares in controls suggests that within-metro resorting is not the primary driver of the results.

7 Mechanisms and Heterogeneity

Section 3 outlined several potential channels through which public housing construction might affect neighborhoods: mechanical compositional channels from adding public housing residents, endogenous sorting responses by private households, and externalities from the build environment or concentrated poverty. In this section, I explore these mechanisms empirically by examining heterogeneity in the neighborhood effects of public housing construction. I present results at $t = 1$ and $t = 3$ to highlight both short/medium-run and long-run effects.

7.1 Mechanical vs Behavioral Effects

I first explore whether effects were purely mechanical. To do so, I examine whether treatment effects vary with the share of tract residents living in public housing in the post-treatment period ($t = 0$). If effects were purely mechanical, we would expect them to scale proportionally with this share. Figure 15 presents treatment effects separately for treated neighborhoods in the bottom, middle, and top terciles in the public housing population share at $t = 0$. For median income and rents, all three terciles show declines of similar magnitudes. This pattern provides suggestive evidence against purely mechanical effects: if income declines simply reflected the mechanical addition of low-income public housing residents, we would expect larger effects in neighborhoods where public housing residents constitute a larger share of the population. The similar magnitudes across terciles are consistent with public housing triggering broader sorting responses in the private housing market, though I cannot definitively rule out mechanical explanations.

For population outcomes, the pattern is less clear. Black and total population increases are larger in neighborhoods with higher public housing shares, consistent with mechanical effects. However, behavioral responses could amplify mechanical population changes, for example, if the arrival of Black public housing residents attracted additional Black in-migration or accelerated white out-migration. The income and rent results suggest that such amplification is plausible.

7.2 Neighborhood Tipping

I next examine whether these behavioral responses operated through racial tipping dynamics. Models of neighborhood tipping suggest that the initial racial composition of a neighborhood may influence whether demographic shocks precipitate broader racial transition (Schelling 1971; Card, Mas, and J. Rothstein 2008). Some historical accounts of public housing describe dynamics consistent with tipping, where public housing construction coincided with a rapid neighborhood racial transition (R. Rothstein 2017; Jackson 1985). To test this tipping hypothesis, I divide treated neighborhoods into three groups based on their baseline Black population share: almost entirely non-Black (less than 1%; $N = 258$), those in a potential "tipping range" (1–12%; $N = 186$), and those with higher initial Black shares (above 12%; $N = 360$). I choose 12% as the upper bound of the tipping range because it corresponds to the top of the threshold estimated by Card, Mas, and J. Rothstein (2008) in 1970. I also analyze nearby neighborhoods using the same cutoffs. I use population by race in levels rather than inverse hyperbolic sine here, since percentage changes in Black/white population will mechanically be larger/smaller in neighborhoods with low/high initial Black population shares.

Figure 16 presents the results, which show that the effects of public housing on racial composition depend strikingly on the initial neighborhood context. In treated neighborhoods in the tipping range, I find large increases in Black population and large declines in white population in both treated and nearby neighborhoods, consistent with racial tipping triggering white flight. In contrast, treated neighborhoods that were almost entirely non-Black at baseline saw moderate increases in Black population but no significant change in white population, while treated neighborhoods with high initial Black shares saw no change in Black population and potentially small increases in white population.

Nearby neighborhoods experienced similar patterns: nearby neighborhoods in the tipping range also saw sizable declines in white population, providing evidence that in this range, public housing construction triggered broader white flight dynamics beyond the treated neighborhoods themselves. These nearby neighborhoods also saw increases in Black population, though notably smaller than in treated neighborhoods. Overall, these results suggest that racial tipping dynamics played a significant role in shaping the neighborhood effects of public housing construction. These results complement findings on modern affordable housing programs that the neighborhood effects of new affordable housing construction depend on initial neighborhood characteristics (Diamond and McQuade 2019).

7.3 Heterogeneity by Project Characteristics

Next, I explore whether the effects of public housing varied based on the type of projects. As discussed in Section 3, the physical structure of public housing could create built-environment externalities, with high-rise towers potentially serving as local disamenities through their physical design or by fostering crime and social disorder (Jacobs 1961; Newman 1972). These concerns were codified in 1977, when HUD prohibited the construction of new high-rise public housing for families. Given these concerns, I test whether effects differed by building type, classifying projects as high-rise or not. Building characteristics are available from the digitized reports for Chicago and New York City, and for a subset of projects in other cities from the modern HUD administrative dataset. I define high-rise projects as those designated "elevator" buildings in Chicago and in the NGDA, and as those with greater than seven floors in New York City, following HUD's modern definition.

Figure 17 presents the results by building type. With the caveat that these building characteristics are incomplete, I find little evidence that high-rise projects had systematically worse neighborhood effects than low-rise projects. If anything, non-high-rise projects appear to have had somewhat larger negative effects on median income and rents, though the differences are not statistically significant. The spillover results are similarly unexpected: nearby neighborhoods around non-high-rise projects experienced significant white population declines and Black share increases, while high-rise projects show no significant spillover effects on racial composition. These results suggest that building design may not have been a primary driver of the neighborhood effects of public housing construction. However, a caveat is in order: Because, outside of New York City and Chicago, I only have building type information for a subset of projects that existed in 2023, these estimates may be missing some of the most distressed high-rise projects demolished in prior decades. Going forward, I plan to collect more complete building type data to provide a more definitive test of this hypothesis.

7.4 Additional Heterogeneity Analyses

Finally, I explore several additional dimensions of heterogeneity. I first examine whether effects differed in New York City and other cities in my sample. The New York City Housing Authority (NYCHA) was, and remains, the largest housing authority in the country, and it has often been thought of as better managed and more successful than other American public housing programs (Bloom 2008; Hunt 2018). Figure 18 shows results separately for New York City and for other cities. The results are somewhat nuanced. Comparing the long-run effects, treated neighborhoods in New York City experienced smaller and less statistically significant declines in median income and rents compared to those in other cities. However, white flight patterns are more pronounced in New York City, with nearby neighborhoods around New York City projects experiencing much larger and more significant white population declines than in other cities. Overall, these results paint a mixed picture, and suggest that the neighborhood effects of public housing may not neatly correspond with the perceived success of local housing authorities.

I also test whether the effects of the projects varied depending on when they were built. These results are shown in Figure 19. I find that the effects of public housing differed substantially based on when projects were built, particularly for population and racial composition. Early-period projects led to larger and more persistent Black population increases in treated neighborhoods, while the effects for later-period projects were smaller and faded over time. More strikingly, nearby neighborhoods around projects built in the early period saw notable long-run white outflows, while nearby neighborhoods around late-period projects saw no significant white population changes. On the other hand, the effects on median incomes and rents were similar across the two periods. One possible explanation for these differences is that urban neighborhoods had lower Black population shares in the 1940s and 1950s than later, and so public housing construction in that period may have been more likely to trigger the tipping dynamics shown in Section 7.2.

Finally, I test whether these effects differed based on whether the neighborhoods were also affected by the urban renewal program. This analysis both highlights potential interactions between the two programs and addresses concerns that my baseline estimates may be conflated with the effects of urban renewal. Figure 20 presents the results, separately estimating Equation 3 for neighborhoods that were urban renewal tracts and those that were not. The main finding is that the neighborhood effects of public housing construction seem largely to be concentrated in non-urban renewal areas. Non-urban renewal tracts show the large Black population increases and income declines seen in the main results, while urban renewal tracts show no significant demographic changes. This pattern is consistent with LaVoice (2024), who finds that urban renewal led to declines in Black population shares. In neighborhoods where both programs operated, the displacement effects of urban renewal may have counteracted the demographic changes associated with public housing construction.

8 Did Public Housing Create Low-Opportunity Neighborhoods

I have shown that public housing had significant and persistent effects on neighborhood population, racial and economic composition, and housing markets. I next ask whether these changes translated to differences in long-run neighborhood opportunity for children who grew up in these neighborhoods.

I explore this question by merging tract-level data from the Opportunity Atlas (Chetty et al. 2018). These data include measures of long-run upward mobility and incarceration for children born from 1978 to 1983 whose parents were at the 25th percentile of the national income distribution. I will refer to these children as "low-income children". In particular, I use tract-level measures of the mean household income rank in adulthood in 2014-2015 and the share of these children who were incarcerated as of April 1st, 2010, at the 2010 Census tract level. I concord these data to 1950 tract boundaries using my tract crosswalks, along with population counts of relevant children in each tract from the Opportunity Atlas data. I then match these data to the matched pairs of public housing tracts, nearby tracts, and control tracts used in Section 6.

I then estimate the following regression model separately for public housing tracts and nearby tracts:

$$Y_{im} = \beta \text{Treated}_i + X'_{i,1980}\gamma + \mu_m + \epsilon_{im} \tag{4}$$

where Y_i denotes the Opportunity Atlas outcome of interest, either the mean income rank or the incarceration rate for low-income children, in tract i, μ_m denotes a matched-pair fixed effect, and $X_{i,1980}$ is a vector of tract characteristics in 1980 (Black share, median income, total population, unemployment rate, and median rent). The variable Treated_i is an indicator for whether tract i is a public housing tract (or a nearby tract). The coefficient β captures the average difference in outcome between each treated tract and its matched control tract, conditional on 1980 neighborhood characteristics. Since most public housing projects were built before 1980, the 1980 characteristics are themselves affected by public housing and should therefore be considered "bad controls" in the usual causal inference sense. However, including these controls allows me to distinguish between effects operating through observable neighborhood composition versus the presence of public housing itself.

Tables 6 and 7 present results for public housing tracts and nearby tracts, respectively. I estimate specifications both without controls (Columns 1 and 3) and with controls for 1980 neighborhood characteristics (Columns 2 and 4).

Children who lived in public housing tracts experienced significantly worse outcomes: a 1.8 percentage point lower income rank and a 0.6 percentage point higher incarceration rate compared to matched controls (Table 6, Columns 1 and 3). About a quarter of this effect persists after controlling for 1980 neighborhood characteristics, such as Black share and median income (Columns 2 and 4), suggesting that public housing had effects on neighborhood upward mobility and incarceration rates even beyond simply changing observable demographics.

By contrast, the apparent spillover effects on nearby tracts appear to reflect selection rather than true spillovers. While nearby tracts initially show worse outcomes (Table 7, Columns 1 and 3), these differences completely disappear when controlling for 1980 neighborhood characteristics (Columns 2 and 4). These results imply that neighborhoods closer to public housing experienced worse outcomes simply because they were poorer and more Black by 1980, not because proximity to public housing directly reduced opportunity. The adverse effects of public housing on upward mobility were therefore geographically concentrated within immediate project tracts themselves.

Taken together, these results suggest that public housing shaped neighborhood opportunity through two channels: by altering observable neighborhood composition, and to a lesser extent, through additional effects of the projects themselves. These results are descriptive associations, not causal effects. While the matched-pair fixed effects control for pre-treatment neighborhood differences, families who remained in (or moved to) public housing neighborhoods by the late 1970s may differ systematically from those in control neighborhoods. The 1980 controls are themselves affected by treatment, and children experienced different exposures depending on when their neighborhood's project was built. Nonetheless, the patterns are consistent with the neighborhood effects documented in earlier sections translating into worse long-run outcomes for children.

9 Conclusion

This paper has examined the long-run neighborhood effects of the mid-20th-century U.S. public housing program. Using a newly constructed national dataset of more than 8,000 projects built between 1935 and 1973, linked to a consistent panel of tract-level census data spanning 1930–2010, I documented two central findings. First, public housing projects were systematically targeted toward neighborhoods that were initially poorer, more populated, disproportionately Black, and slated for urban renewal. These patterns confirm the program's slum-clearance origins and the racialized politics of site selection. Second, public housing construction had significant and persistent effects on neighborhood trajectories: treated tracts experienced long-run increases in Black population shares and sustained declines in incomes and rents, with modest spillover effects on nearby neighborhoods including white population decline and income reductions. Importantly, I find evidence consistent with racial tipping dynamics: neighborhoods with moderate baseline Black shares experienced substantial white population outflows following public housing construction, while neighborhoods with higher initial Black shares showed more muted responses.

Taken together, these results suggest that public housing not only reflected but also reinforced existing patterns of segregation and disinvestment, contributing quantitative evidence to longstanding debates about the role of federal policy in creating residential segregation (R. Rothstein 2017; Trounstine 2018; T. D. Logan and Parman 2025). Rather than catalyzing neighborhood improvement as envisioned in the Housing Act of 1949, the program contributed to the concentration of poverty and the persistence of racial segregation in American cities. The historical patterns I document have clear implications for contemporary affordable housing policy. Modern reforms to public housing have emphasized smaller-scale interventions, mixed-income developments, and designs that blend into existing neighborhoods rather than large, segregated projects. My findings help explain the rationale behind these reforms: large concentrations of subsidized housing in neighborhoods undergoing racial transition accelerated segregation and demographic change.

One crucial caveat is that my results do not necessarily imply that public housing failed to benefit individual residents. For many low-income residents, public housing may have provided better housing quality, stability, and affordability than available private-market alternatives. In ongoing work, I am linking individuals to public housing projects using full-count Census data to study the effects on project residents and their neighbors. This individual-level analysis will complement the neighborhood-level findings presented here by directly measuring the benefits and costs experienced by public housing residents themselves, providing a more complete assessment of the program's legacy.

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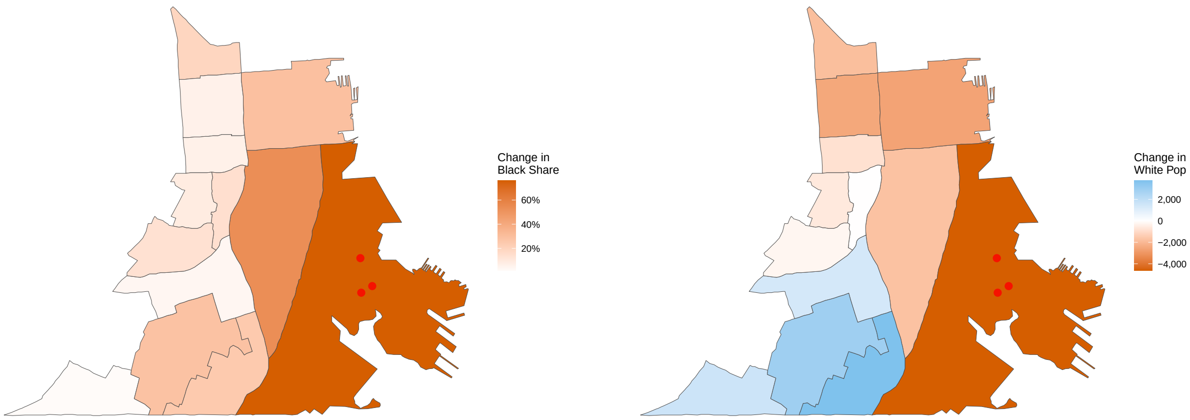
Figures and Tables

Figure 1: Motivating Example: Hunters Point, San Francisco

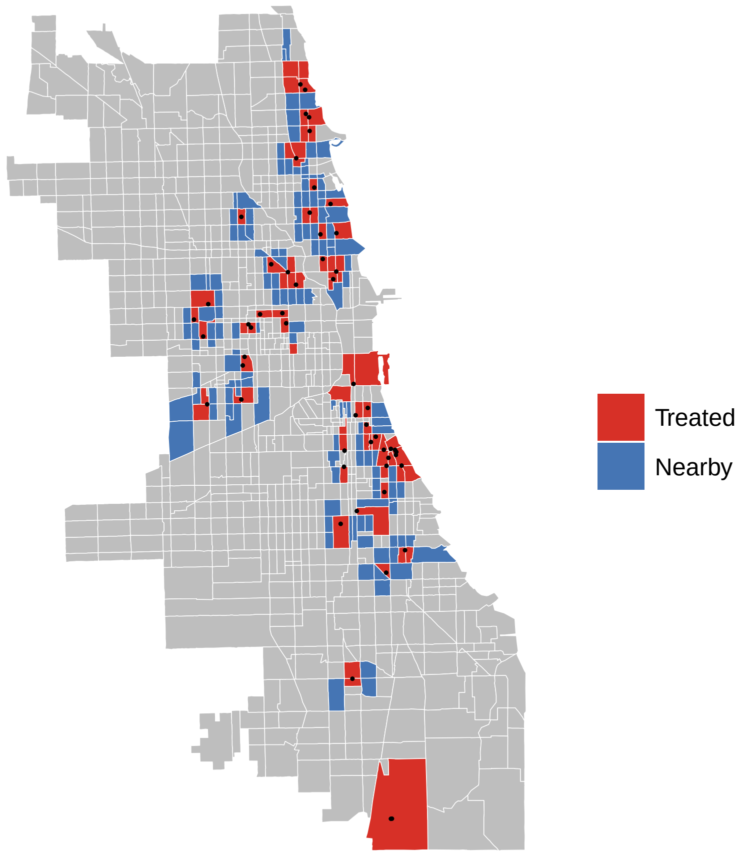
(a) Change in Black Population Share

(b) Change in White Population



Note: These maps show demographic changes from 1940 to 1970 in the census tract containing Hunters Point public housing (built 1942–1944) and surrounding tracts within 2 kilometers. Panel (a) shows the change in Black population share; panel (b) shows the change in white population. Red dots indicate public housing project locations. See Section 2 for historical context.

Figure 2: Public Housing Projects and Spillover Areas: Chicago



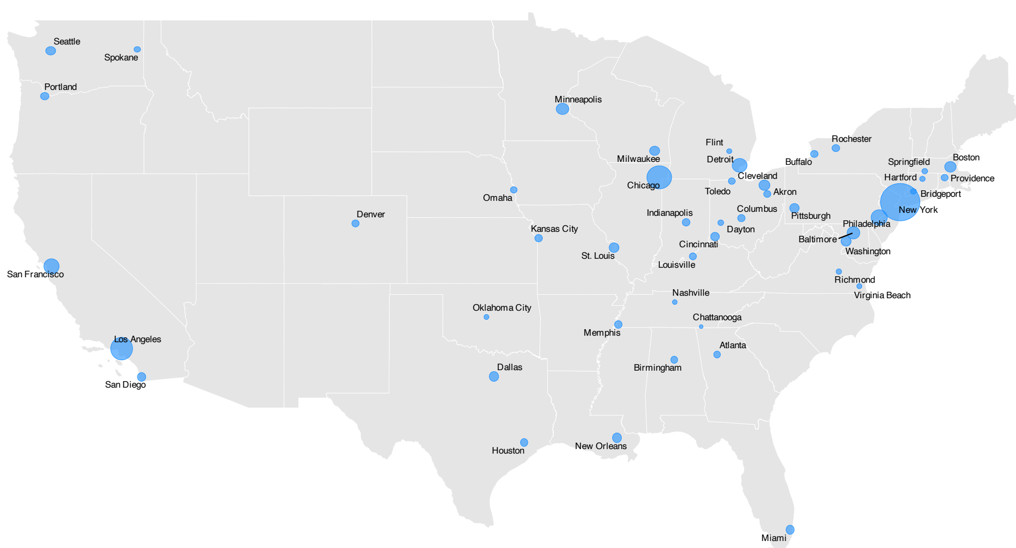
Note: This map shows census tracts in Chicago containing public housing projects built between 1941 and 1973 (red, N=72 tracts) and nearby spillover areas (blue). Spillover areas are defined as census tracts that share a border with a treated tract and are within 1 kilometer of the nearest public housing project. See Section 4 for sample construction details.

Table 1: Sample Attrition

Step	Tracts	Pop. 1940 (M)	CBSAs	Treated Tracts	Projects	Units
Original sample	12,062	38.1	60	1,396	1,540	516,084
Balanced on years (1930-2010)	9,086	34.1	50	1,145	1,207	466,303
Complete variables	8,596	33	50	1,119	1,178	451,811
Exclude treatments ≤1940	8,488	32.4	50	1,011	1,067	394,950
Drop CBSAs <30 tracts	8,431	32.1	47	996	1,039	390,759
Drop population outliers	6,507	27.5	47	820	811	299,900

Notes: This table shows the impact of sequential sample restrictions on the analysis sample. Columns report the number of census tracts, 1940 population (in millions), Core-Based Statistical Areas (CBSAs), treated tracts (those receiving public housing), public housing projects, and total housing units at each stage. The final balanced sample represents approximately 21% of the 1940 U.S. population. See Section 4 for detailed discussion of each restriction.

Figure 3: Geographic Coverage of CBSAs in Analysis Sample



Note: This map shows the 47 Core-Based Statistical Areas (CBSAs) included in the balanced panel analysis, containing 6,507 census tracts and 820 treated tracts. Shaded areas represent metropolitan areas meeting the sample selection criteria: CBSAs with at least one public housing project built between 1941 and 1973, at least 30 total tracts, and complete census tract data for all study years (1930–2010). See Section 4 for sample construction details.

Table 2: Comparison of Public Housing and Non-Public Housing Neighborhoods (1940 Baseline)

Variable	Untreated	Treated	Std. Diff.
N (Tracts)	5672	820	
Total Population (asinh)	8.83 (0.64)	9.09 (0.60)	0.422***
Black Share	0.06 (0.17)	0.16 (0.28)	0.440***
Black Population (asinh)	3.23 (2.56)	4.89 (2.85)	0.613***
Median Income (asinh)	3.61 (0.30)	3.42 (0.31)	0.608***
Median Rent (asinh)	6.59 (0.42)	6.39 (0.43)	0.459***
Unemployment Rate	0.11 (0.05)	0.14 (0.06)	0.524***
Labor Force Participation Rate	0.56 (0.05)	0.56 (0.05)	0.141***
Distance from CBD (asinh)	8.05 (2.85)	6.99 (3.42)	0.336***
Share Needing Major Repairs	0.07 (0.09)	0.11 (0.11)	0.394***

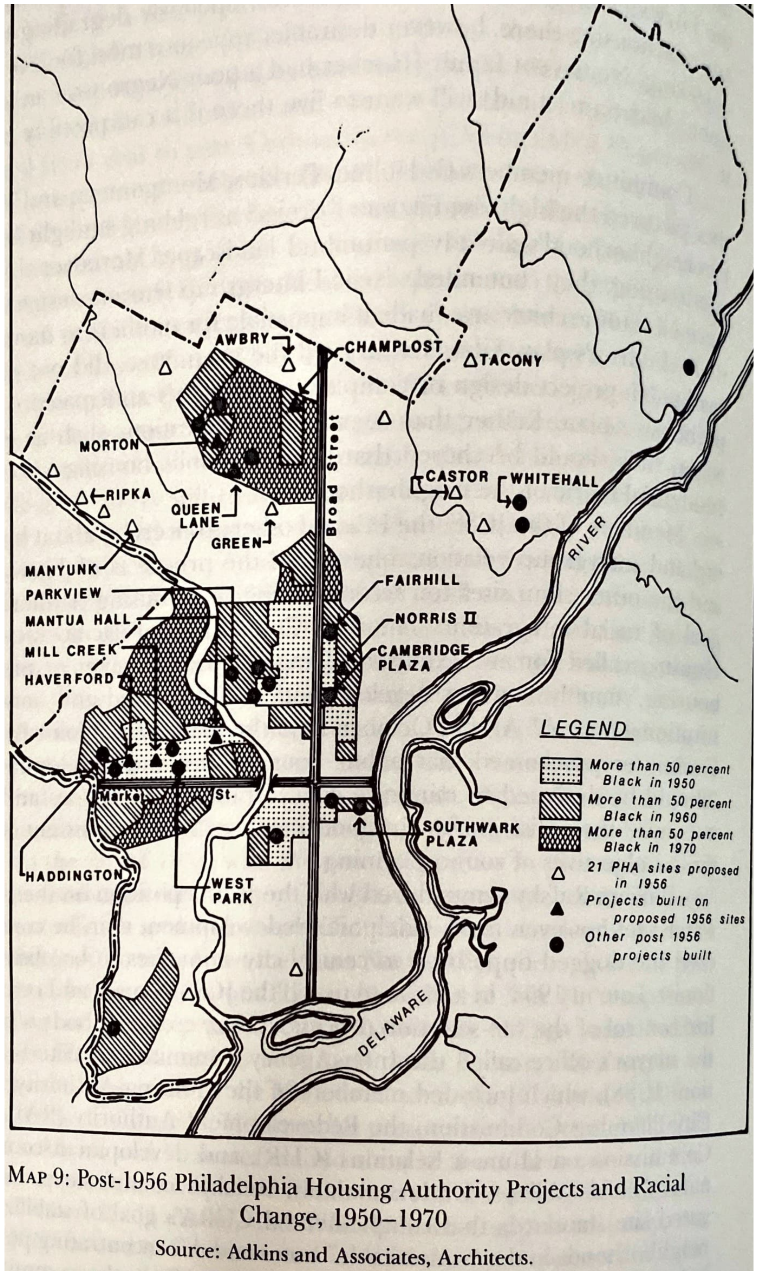
Notes: This table compares 1940 baseline characteristics between census tracts that eventually received public housing projects between 1941 and 1973 (Treated, N=820) and those that did not (Control, N=5,687). Asinh denotes the inverse hyperbolic sine transformation. Standard deviations are shown in parentheses. The Std. Diff. column reports standardized mean differences (difference in means divided by pooled standard deviation). Statistical significance from two-sample t-tests: *p<0.1, **p<0.05, ***p<0.01. See Section 5 for analysis of site selection patterns.

Table 3: Site Selection: Predicting Public Housing Placement from 1940 Neighborhood Characteristics

	(1)	(2)	(3)	(4)
Black Share	0.211***	0.190***	0.185***	0.176***
	(0.041)	(0.041)	(0.044)	(0.040)
Asinh Total Population	0.039***	0.039***	0.042***	0.037***
	(0.010)	(0.009)	(0.010)	(0.009)
Asinh Median Income	-0.106***	-0.080***	-0.076***	-0.067***
	(0.020)	(0.020)	(0.020)	(0.020)
Asinh Median Rent	-0.075***	-0.030	-0.021	-0.025
	(0.018)	(0.028)	(0.030)	(0.028)
Pct Graduated HS		0.091	0.098	0.060
		(0.060)	(0.064)	(0.060)
Unemployment Rate		0.851***	0.851***	0.805***
		(0.180)	(0.185)	(0.177)
LFP Rate		-0.356***	-0.301**	-0.358***
		(0.128)	(0.132)	(0.126)
Redlined (HOLC)		0.027	0.029*	0.024
		(0.017)	(0.017)	(0.017)
Asinh Dist. from CBD		0.000	0.001	0.001
		(0.003)	(0.003)	(0.003)
CBD Indicator		-0.024	-0.030	-0.033
		(0.031)	(0.033)	(0.032)
Share Needing Major Repairs			0.105	
			(0.076)	
Asinh Dist. to Highway				0.008
				(0.009)
Urban Renewal Area				0.089***
				(0.024)
Num.Obs.	6507	6507	6170	6507
R2	0.111	0.119	0.118	0.125
R2 Adj.	0.103	0.110	0.110	0.115
County fixed effects	Yes	Yes	Yes	Yes

Notes: This table reports linear probability model estimates of the relationship between 1940 neighborhood characteristics and public housing placement. The dependent variable equals 1 if a census tract received a public housing project between 1941 and 1973 (N=820) and 0 otherwise (N=5,687). The baseline probability of treatment is 12.6%. All specifications include county fixed effects. Standard errors in parentheses are adjusted for spatial correlation following Conley (1999) with a 2-kilometer cutoff. Columns progressively add controls: (1) core demographics; (2) additional neighborhood characteristics; (3) housing conditions; (4)–(5) urban renewal and highway exposure. Statistical significance: *p<0.1, **p<0.05, ***p<0.01. See Section 5 for interpretation.

Figure 4: Historical Map of Proposed and Actual Philadelphia Public Housing Sites, 1956



Note: This map shows the distribution of proposed and actual public housing sites in Philadelphia. Following the 1954 Housing Act, the Philadelphia Housing Authority proposed 21 projects across the city, but many faced significant backlash from white communities and were eventually cancelled. The proposed-but-not-built sites had systematically different baseline characteristics from actual sites (particularly lower Black population shares), illustrating both the political dynamics of site selection and why such sites would not serve as valid counterfactuals in a research design. Source: Bauman (1987), georeferenced by author.

Table 4: 1940 Neighborhood Characteristics: Proposed vs Actual Public Housing Locations in Philadelphia

Variable	Actual Public Housing	Proposed Only	Std. Diff.
Black Share	0.28 (0.29)	0.01 (0.02)	1.310***
Asinh Total Population	9.16 (1.27)	7.98 (1.78)	0.762**
Asinh Median Income	3.47 (0.30)	3.52 (0.28)	0.148
Unemployment Rate	0.21 (0.10)	0.14 (0.05)	0.897**
Asinh Median Rent	6.41 (0.32)	6.45 (0.40)	0.102
Asinh Distance from CBD	8.31 (2.04)	9.75 (0.26)	0.992**
Redlined (HOLC)	0.40 (0.50)	0.08 (0.29)	0.791**
Urban Renewal Area	0.34 (0.48)	0.17 (0.39)	0.398

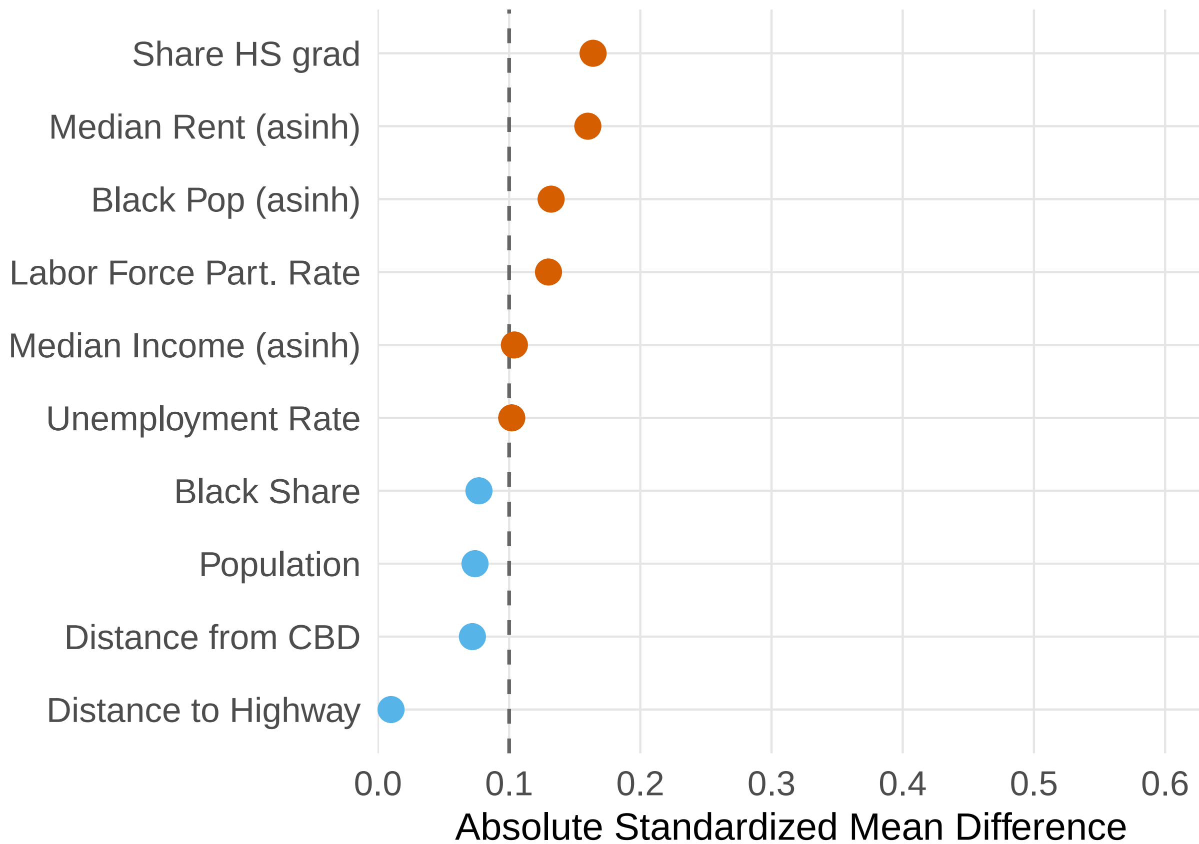
Notes: This table compares 1940 neighborhood characteristics between proposed-but-not-built public housing sites (N=12) identified from a 1956 Philadelphia Housing Authority map (Bauman (1987)) and actually-built sites (N=47) in Philadelphia. The first two columns show means with standard deviations in parentheses. The Std. Diff. column reports standardized mean differences. The large differences, particularly in Black population share, illustrate how political opposition in white neighborhoods shaped site selection, and why proposed-but-rejected sites would not serve as valid counterfactuals. Statistical significance: *p<0.1, **p<0.05, ***p<0.01. See Section 5.2 for discussion.

Table 5: Project Demographics: Predicting Racial Composition of Public Housing from Baseline Neighborhood Characteristics

	(1)	(2)	(3)
(Intercept)	1.554***	1.774***	
	(0.172)	(0.338)	
Black Share	0.435***	0.441***	0.228***
	(0.062)	(0.070)	(0.059)
Asinh Median Income	-0.292***	-0.253***	-0.229***
	(0.042)	(0.056)	(0.058)
Asinh Median Rent		-0.149**	-0.098
		(0.071)	(0.061)
Asinh Population		0.045*	0.052***
		(0.025)	(0.019)
Unemployment Rate		-0.188	0.309**
		(0.182)	(0.125)
LFP Rate		0.242	0.401*
		(0.238)	(0.238)
Asinh Dist. to CBD		0.011*	0.019***
		(0.006)	(0.005)
CBD		-0.075	-0.115*
		(0.071)	(0.066)
Asinh Dist. to Highway		-0.011	-0.045
		(0.032)	(0.034)
Redlined (HOLC)		-0.032	0.024
		(0.054)	(0.050)
Urban Renewal Area		-0.009	0.016
		(0.045)	(0.030)
N	759	759	759
R^2	0.356	0.384	0.581
Adj. R^2	0.354	0.375	0.544
County FE	No	No	Yes

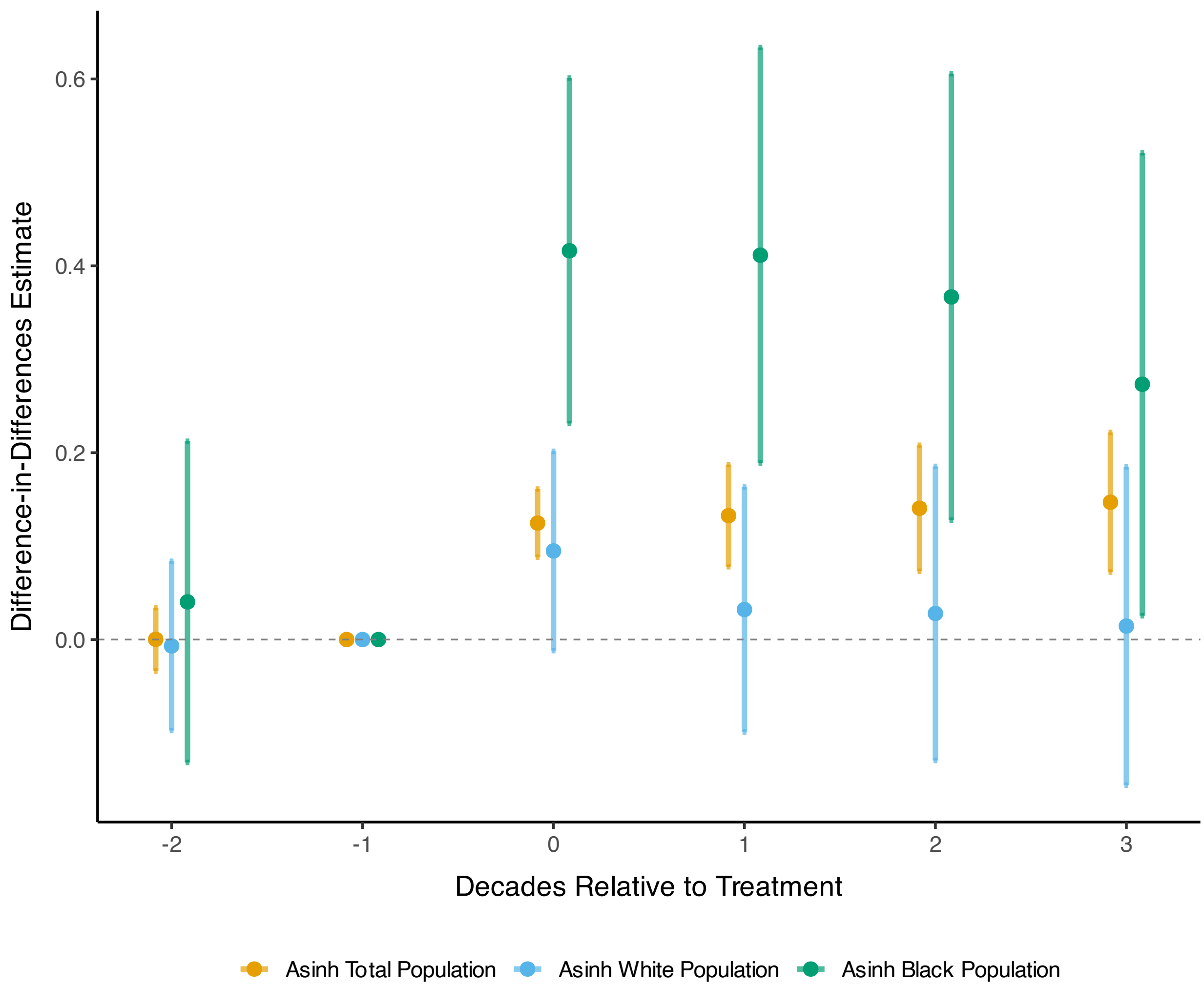
Notes: This table reports OLS estimates of the relationship between baseline neighborhood characteristics (measured at $t-1$, the decade before construction) and project racial composition. The sample includes treated tracts with available project demographic data. The dependent variable is the Black population share within the public housing project, measured from 1970s administrative data (1977 Picture of Subsidized Households for most projects, 1971 data for NYC, 1973 annual reports for Chicago). Column (1) includes baseline Black share and income only; Column (2) adds neighborhood controls; Column (3) adds county fixed effects. Standard errors clustered at the county level in parentheses. Statistical significance: *p<0.1, **p<0.05, ***p<0.01. See Section 5.3 for interpretation.

Figure 5: Pre-period Balance: Treated Neighborhoods vs Matched Controls



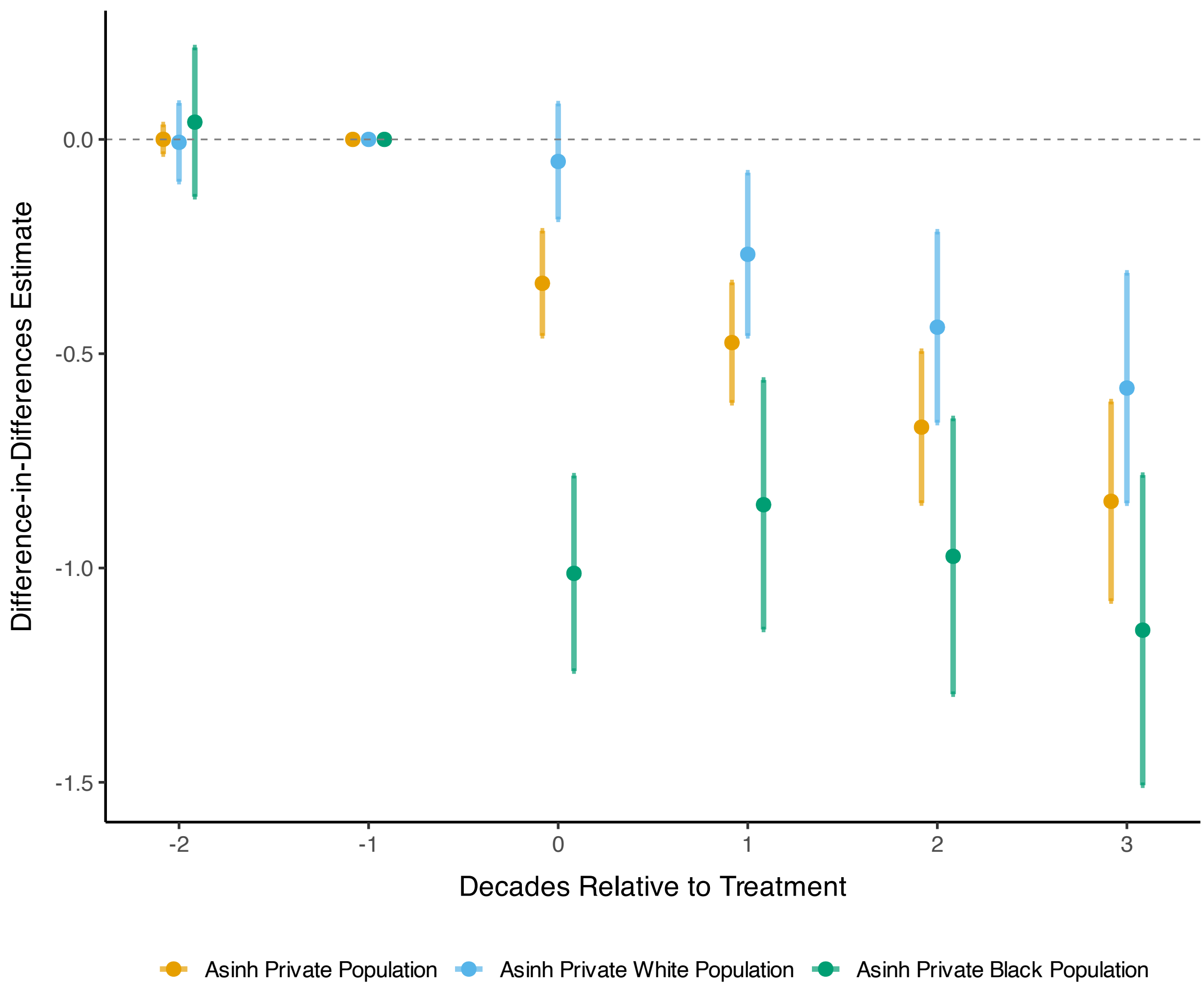
Note: This figure displays standardized mean differences (SMD) between treated neighborhoods (N=804) and their matched controls across baseline covariates. Each point represents the SMD for a covariate measured in the two pre-treatment decades. SMDs near zero indicate successful balance; values above 0.1 (vertical dashed lines) suggest meaningful imbalance. Matching uses propensity scores with exact matching on county and urban renewal status. See Section 6.1 for matching procedure details.

Figure 6: Population and Racial Composition Effects in Treated Neighborhoods



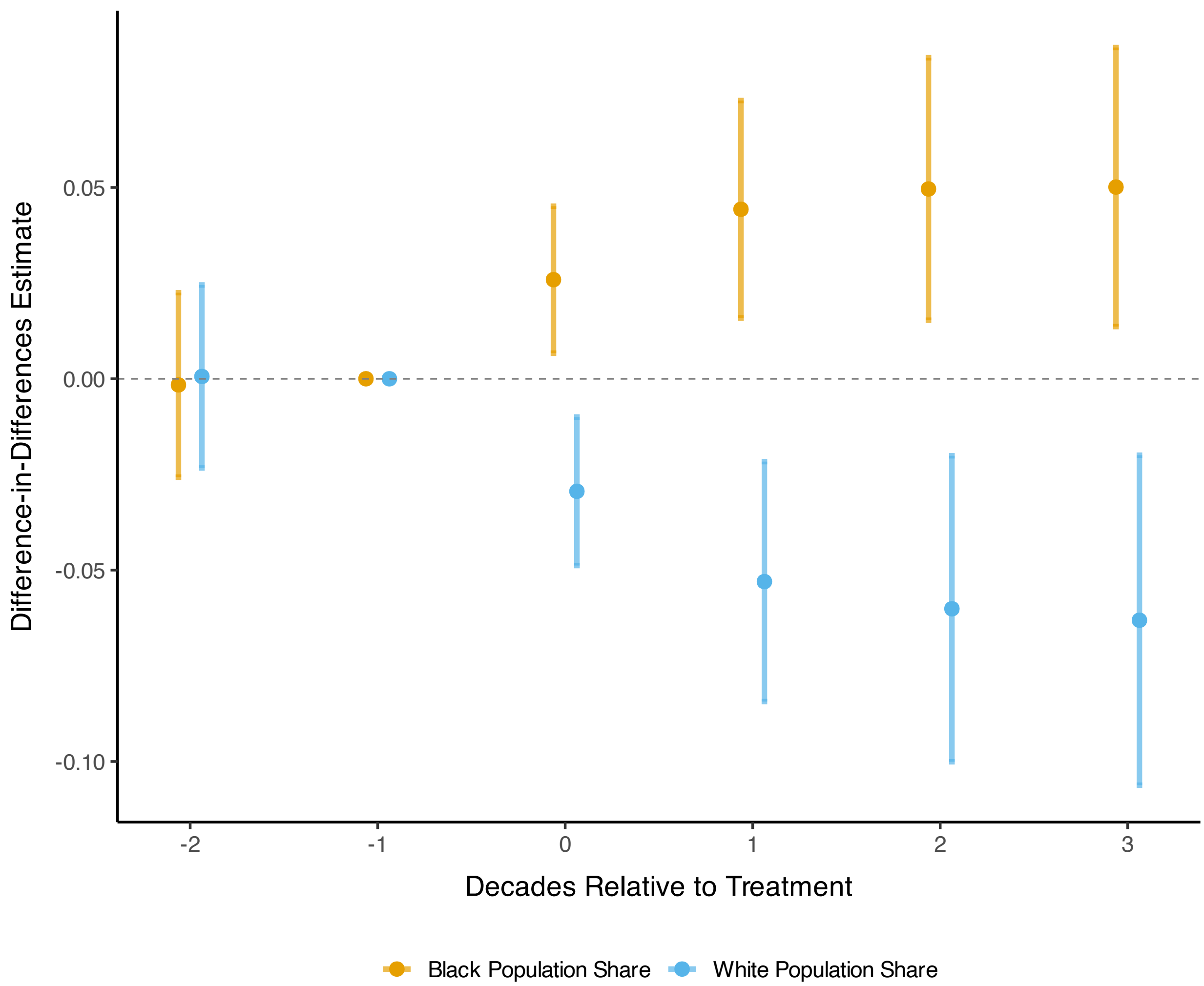
Note: This figure displays event study estimates of public housing effects on population by race in treated neighborhoods (N=804) compared to matched controls, estimated using the stacked difference-in-differences specification in Equation 3. Outcomes are the inverse hyperbolic sine (asinh) of total population, Black population, and white population. Shaded areas represent 95% confidence intervals based on standard errors adjusted for spatial correlation following Conley (1999) with a 2-kilometer cutoff. The reference period is $t = -1$ (one decade before construction). See Section 6.2 for interpretation.

Figure 7: Effects on Estimated Private Population, Treated Neighborhoods



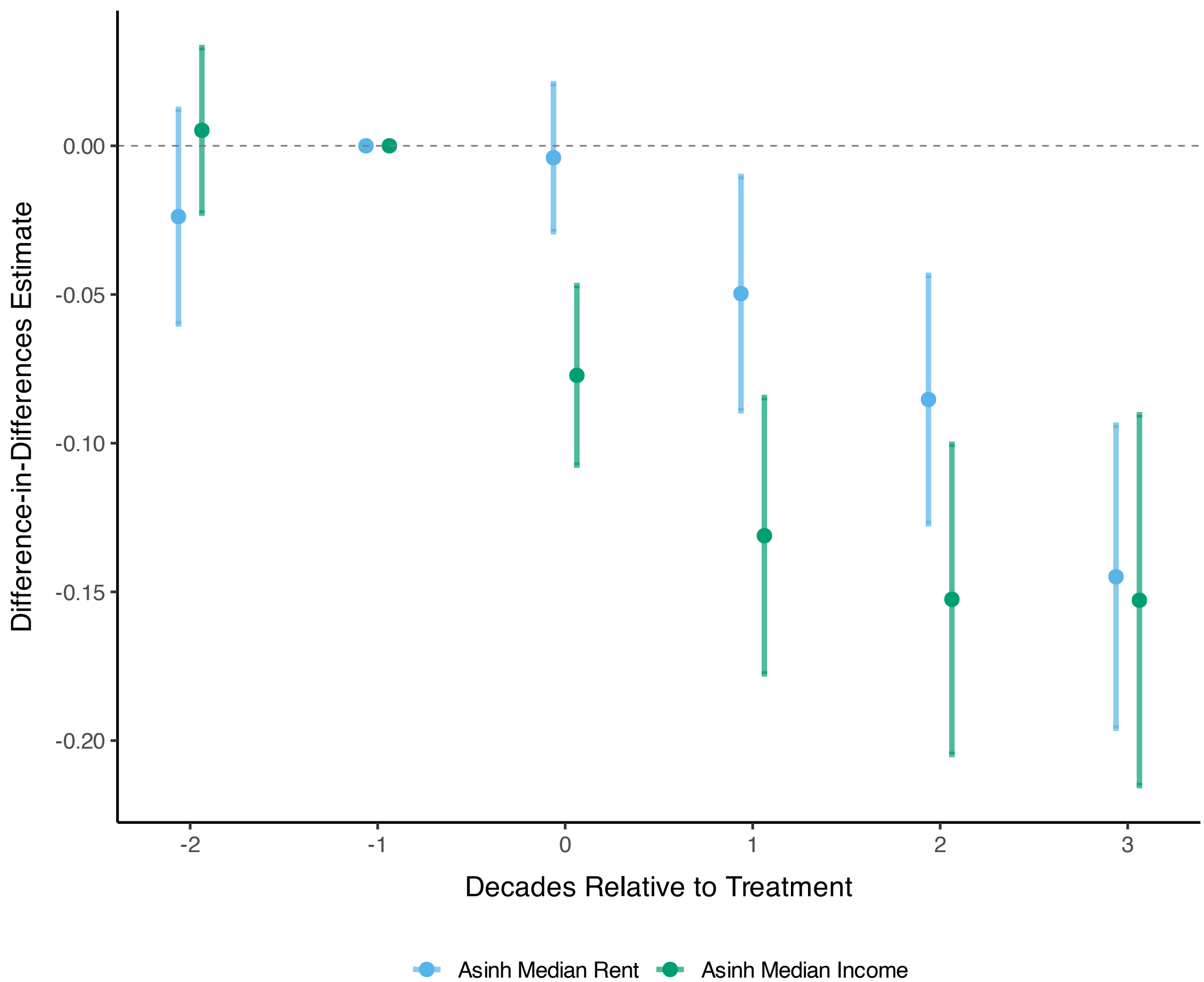
Note: This figure shows event study estimates of public housing effects on the asinh of private (non-public housing) population by race in treated neighborhoods (N=804) compared to matched controls. Private population is estimated by subtracting public housing residents from total tract population; public housing population from 1970s administrative data is set to zero before construction and held constant thereafter. Estimates follow Equation 3. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.2 for interpretation.

Figure 8: Effects on Racial Composition Shares in Treated Neighborhoods



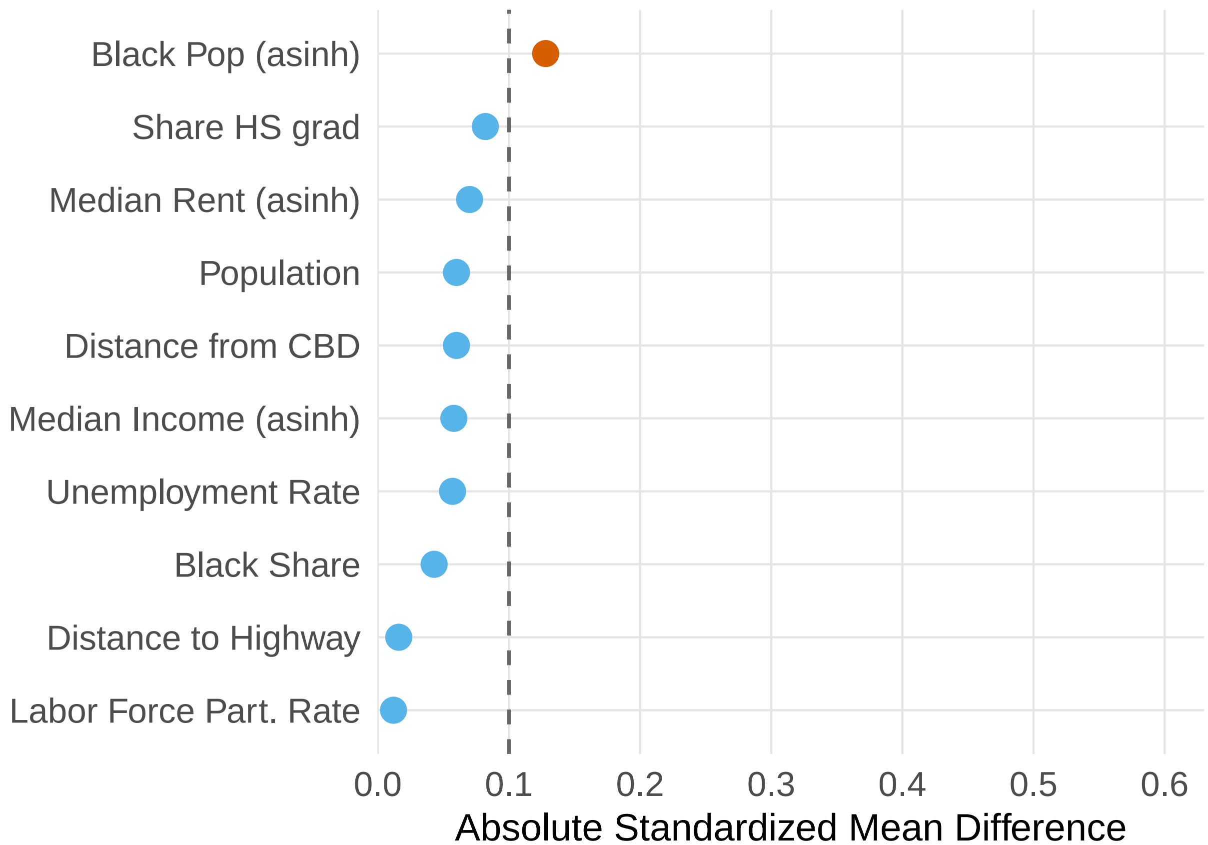
Note: This figure shows event study estimates of public housing effects on racial composition shares in treated neighborhoods (N=804) compared to matched controls, estimated using Equation 3. Coefficients represent percentage point changes in Black and white population shares. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.2 for interpretation.

Figure 9: Economic and Housing Effects in Treated Neighborhoods



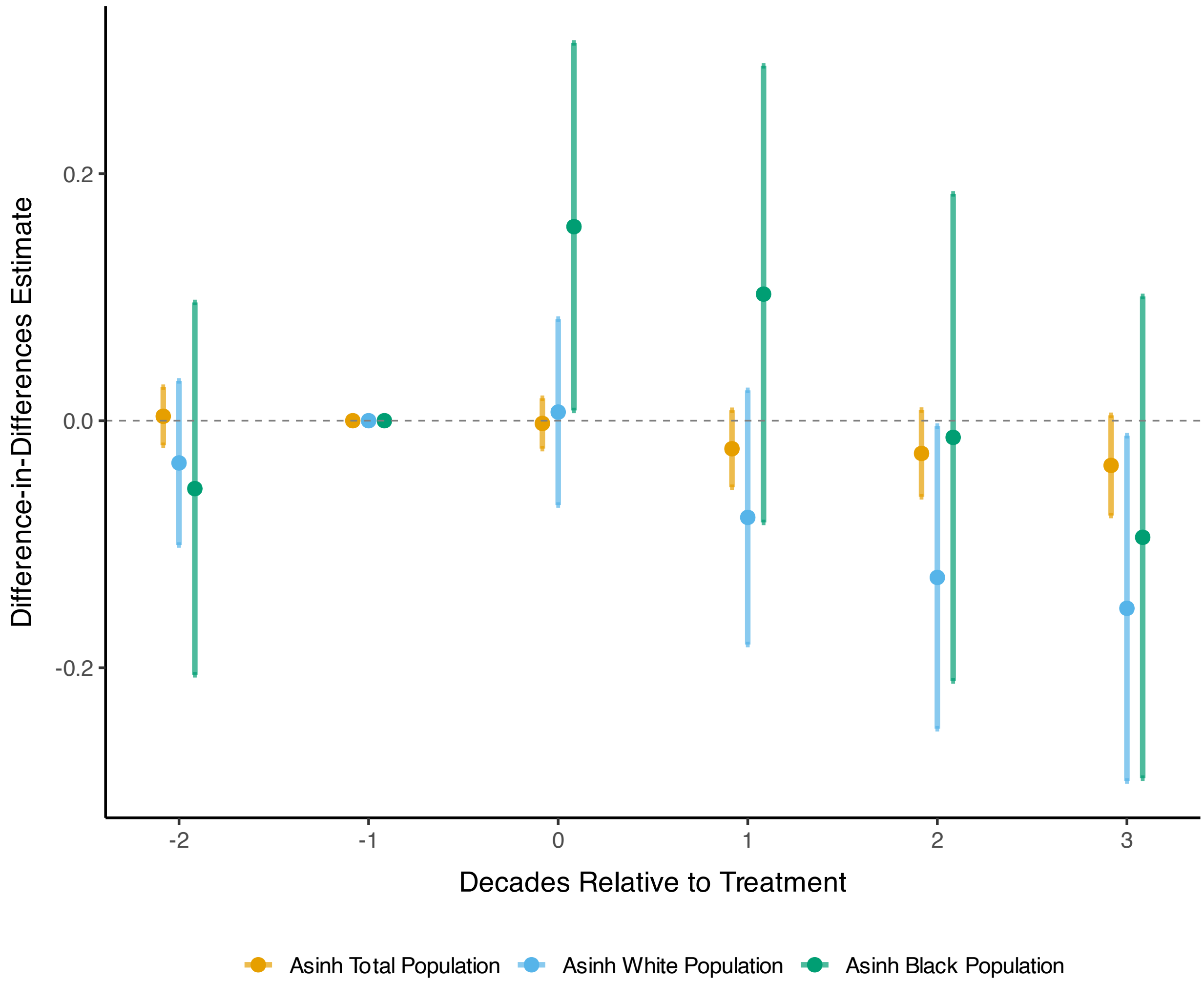
Note: This figure displays event study estimates of public housing effects on the asinh of median rent and median household income in treated neighborhoods (N=804) compared to matched controls, estimated using Equation 3. All monetary values are in 2000 dollars. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.2 for interpretation.

Figure 10: Pre-period Balance: Nearby Neighborhoods vs Matched Controls



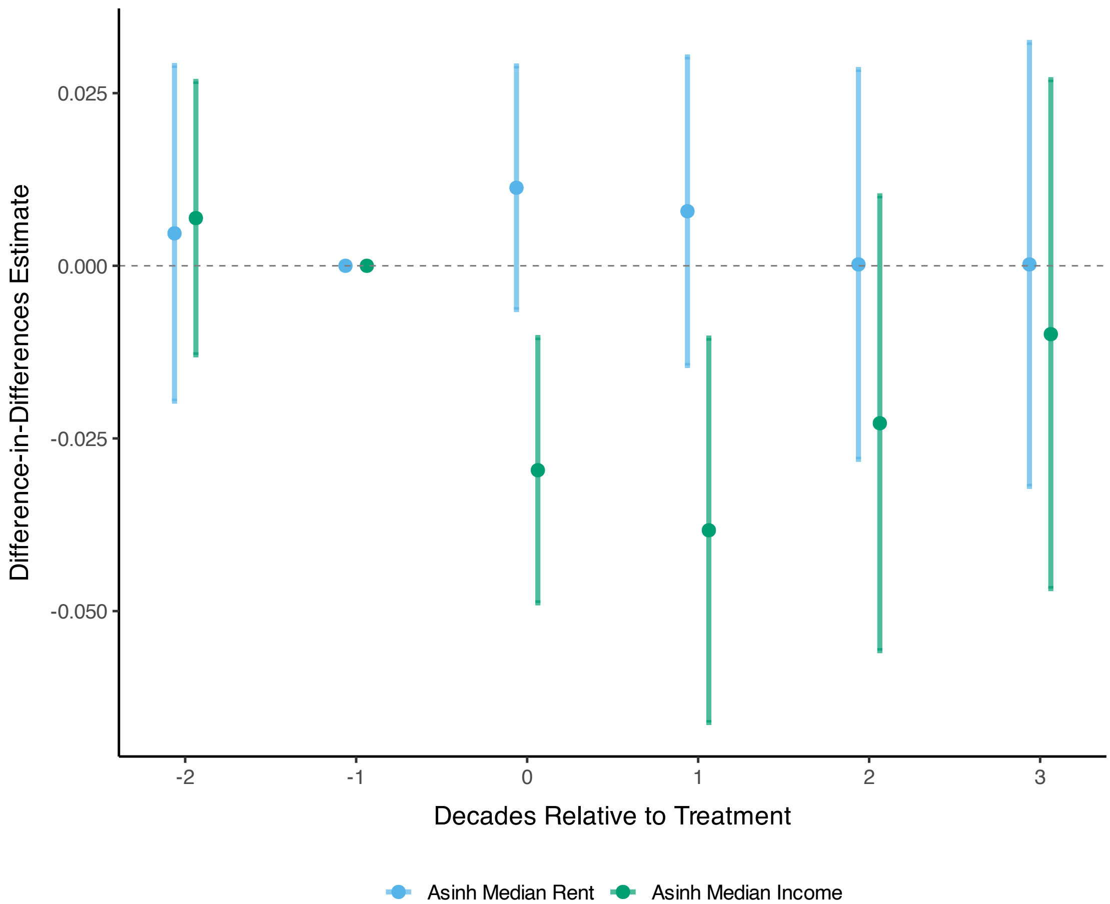
Note: This figure displays standardized mean differences (SMD) between nearby neighborhoods (N=1,414; those sharing a border with treated tracts and within 1km of the nearest project) and their matched controls across baseline covariates. Each point represents the SMD for a covariate measured in the two pre-treatment decades. SMDs near zero indicate successful balance; values above 0.1 (vertical dashed lines) suggest meaningful imbalance. See Section 6.1 for matching procedure.

Figure 11: Spillover Effects: Population and Racial Composition in Nearby Neighborhoods



Note: This figure displays event study estimates of public housing spillover effects on population by race in nearby neighborhoods (N=1,414; those sharing a border with treated tracts and within 1km of projects) compared to matched controls, estimated using Equation 3. Outcomes are the asinh of total, Black, and white population. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.3 for interpretation.

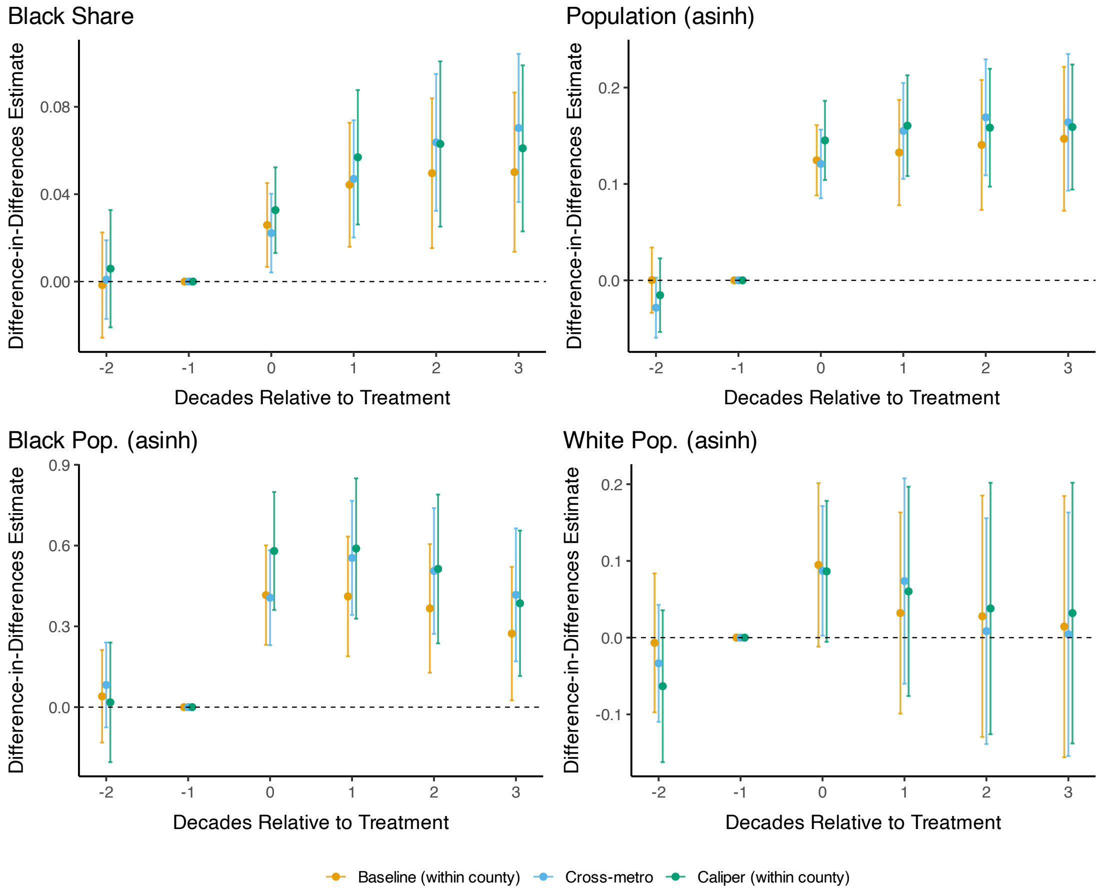
Figure 12: Spillover Effects: Economic and Housing Outcomes in Nearby Neighborhoods



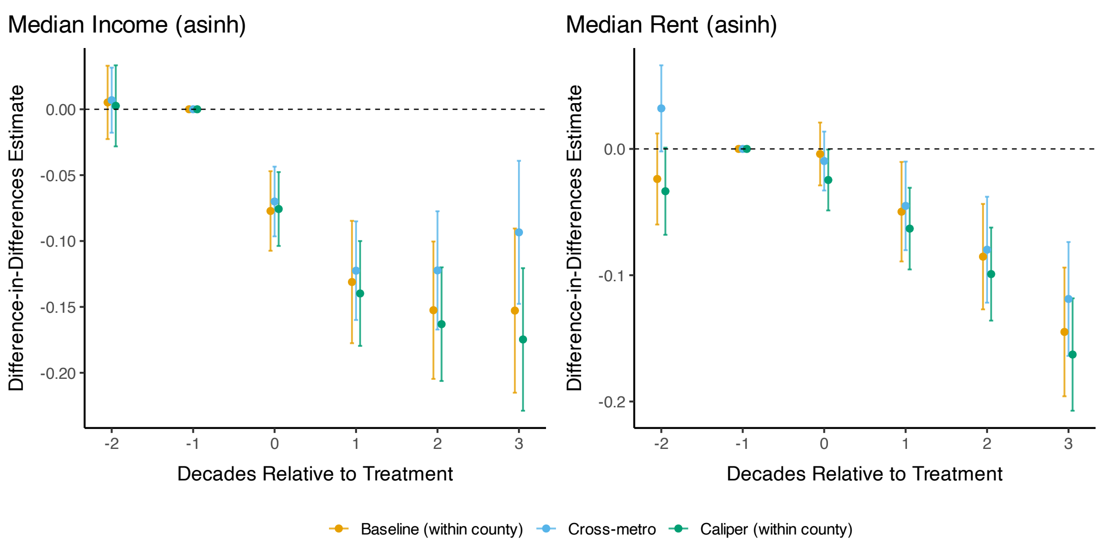
Note: This figure displays event study estimates of public housing spillover effects on the asinh of median rent and median household income in nearby neighborhoods (N=1,414) compared to matched controls, estimated using Equation 3. All monetary values are in 2000 dollars. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.3 for interpretation.

Figure 13: Alternative Matching Specifications: Treated neighborhoods

(a) Demographics



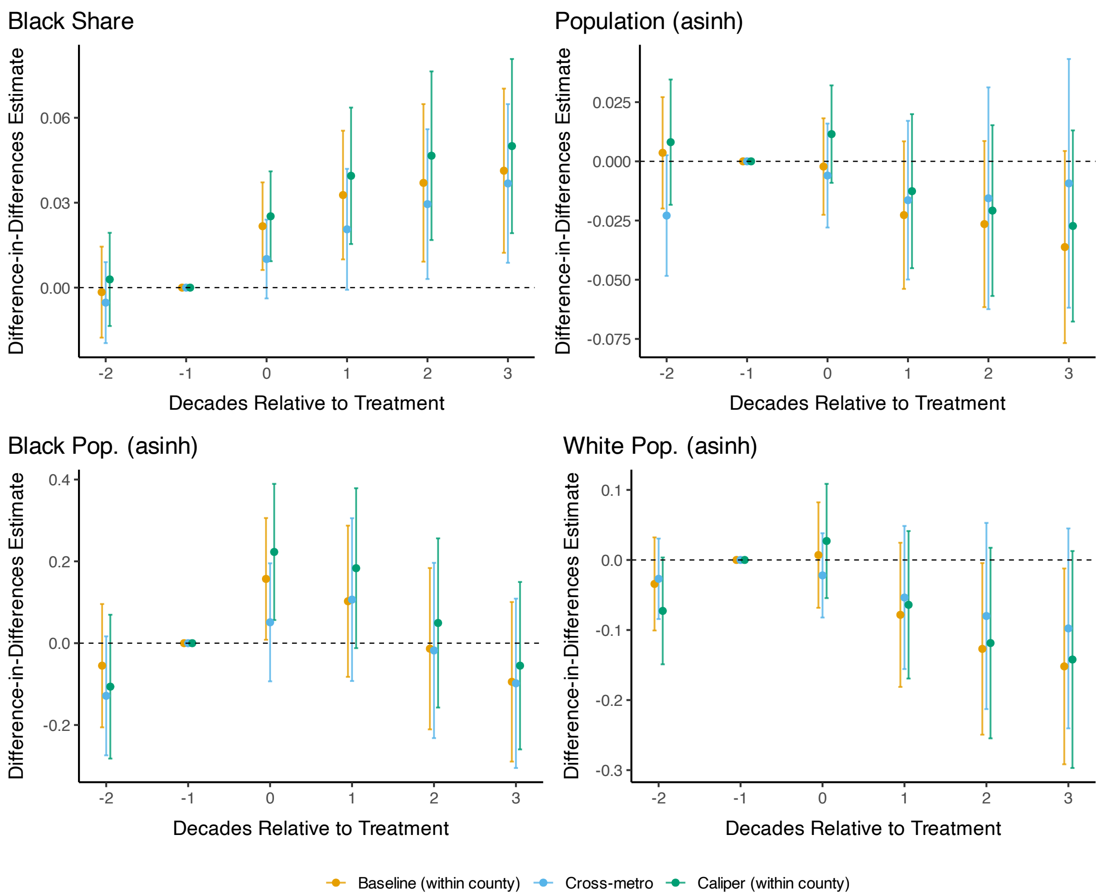
(b) Income & Rent



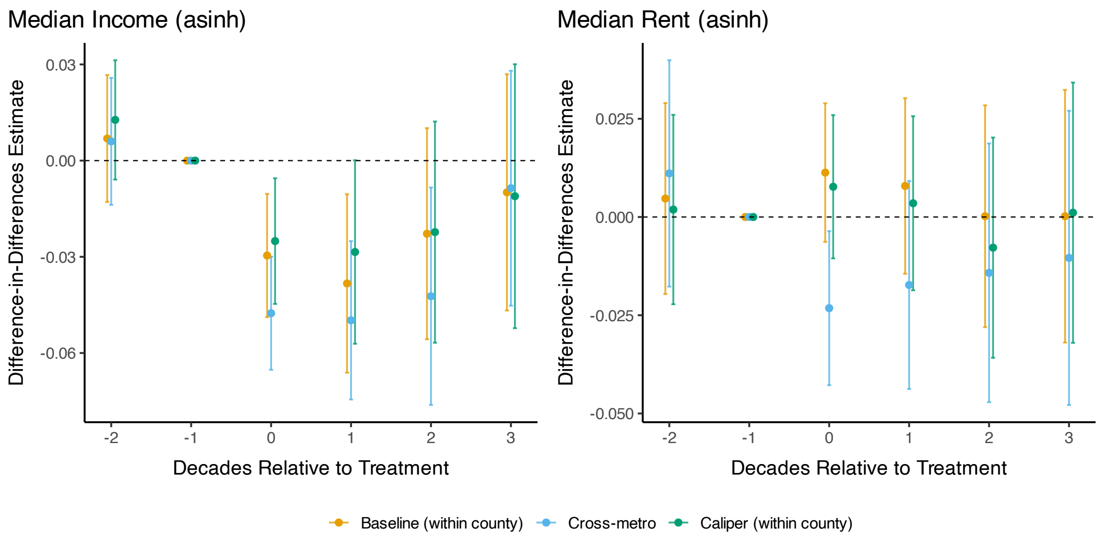
Note: Event study estimates for treated neighborhoods across three matching specifications: (1) Baseline: exact match on county and urban renewal status, N=804; (2) Cross-metro: controls drawn from other CBSAs, exact on urban renewal; (3) Caliper: baseline plus 0.2 SD propensity score caliper. All use 1:1 nearest-neighbor matching with replacement. Shaded areas show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 6.4 for discussion.

Figure 14: Alternative Matching Specifications: Nearby neighborhoods

(a) Demographics



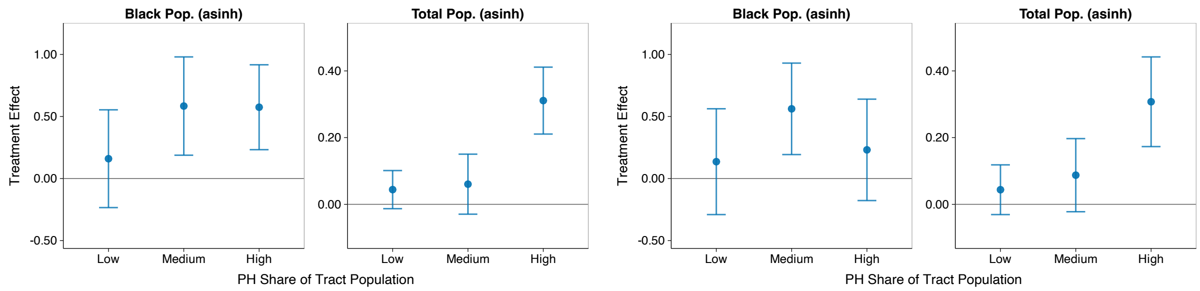
(b) Income & Rent



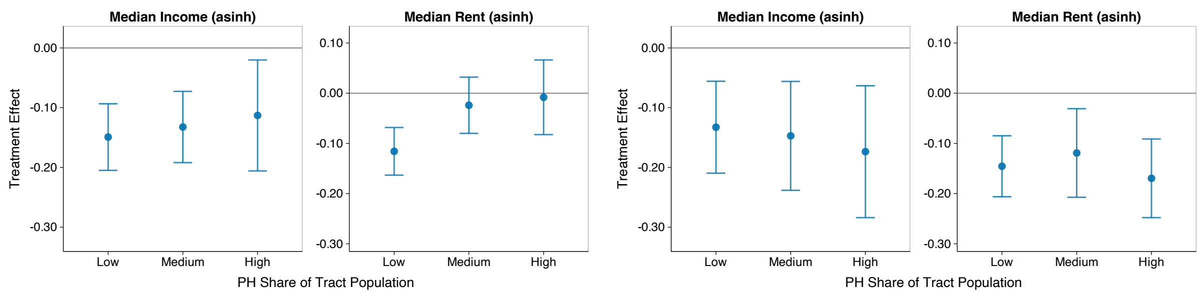
Note: Event study estimates for nearby neighborhoods (N=1,414; border treated tracts, within 1km of projects) across three matching specifications: (1) Baseline: exact match on county and urban renewal; (2) Cross-metro: controls from other CBSAs; (3) Caliper: baseline plus 0.2 SD propensity score caliper. All use 1:1 nearest-neighbor matching with replacement. Shaded areas show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 6.4 for discussion.

Figure 15: Heterogeneity by Public Housing Population Share

(a) Population



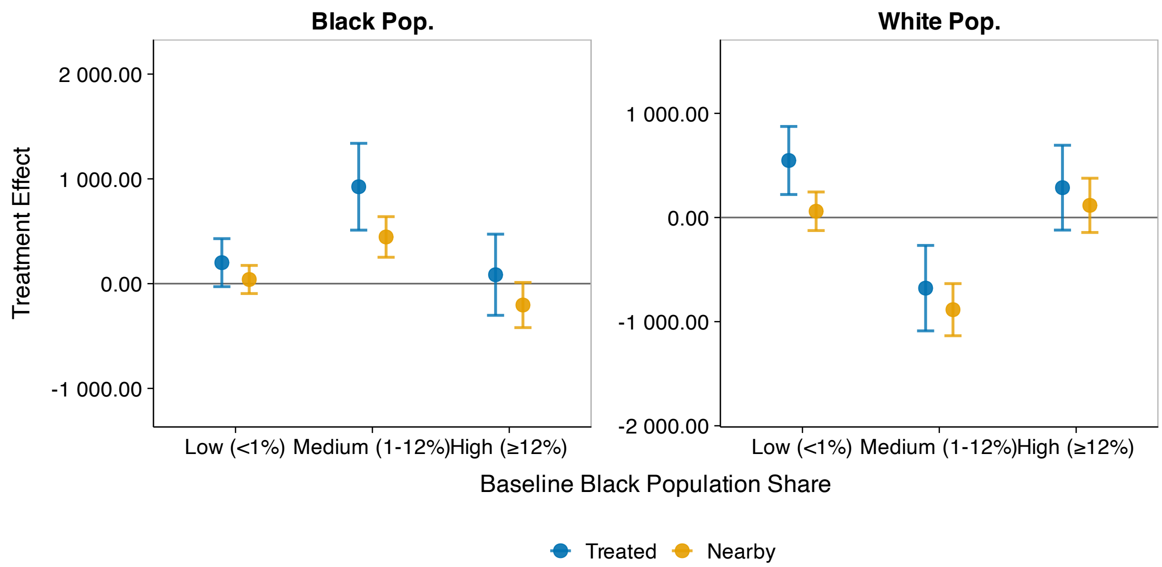
(b) Income and Rent



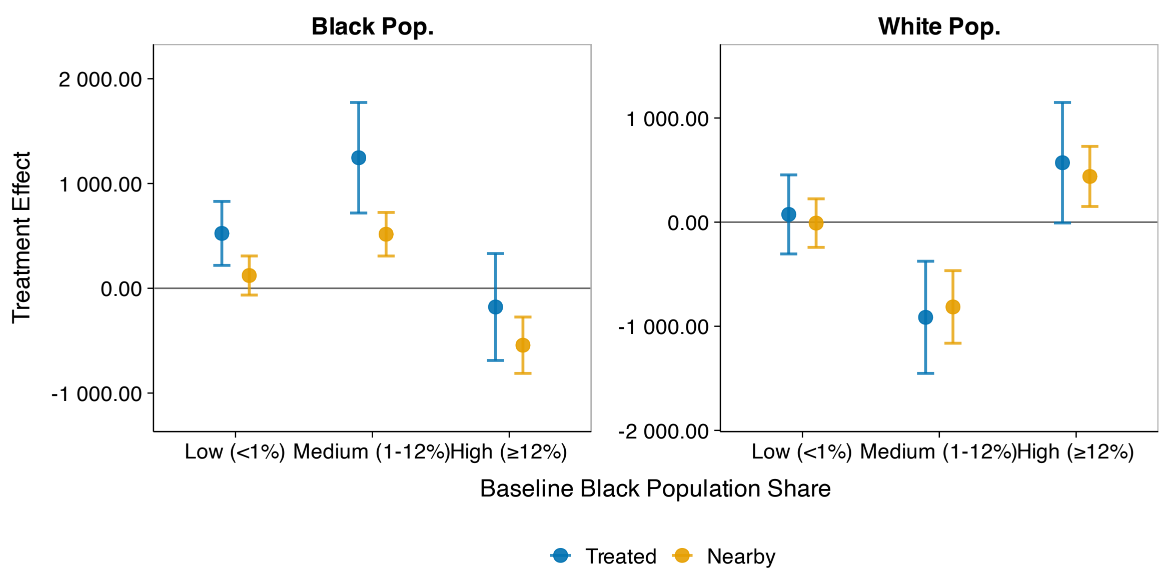
Note: This figure displays heterogeneous treatment effects by the share of tract population living in public housing at $t = 0$, testing whether effects are purely mechanical. Treated neighborhoods (N=804) are divided into terciles. Estimates follow Equation 3. Left panels show effects at $t = 1$ (one decade after construction); right panels at $t = 3$ (three decades after). For income and rent, similar magnitudes across terciles provide suggestive evidence against purely mechanical effects. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7 for interpretation.

Figure 16: Heterogeneity by Baseline Black Population Share

(a) Medium-run Effects ($t = 1$)



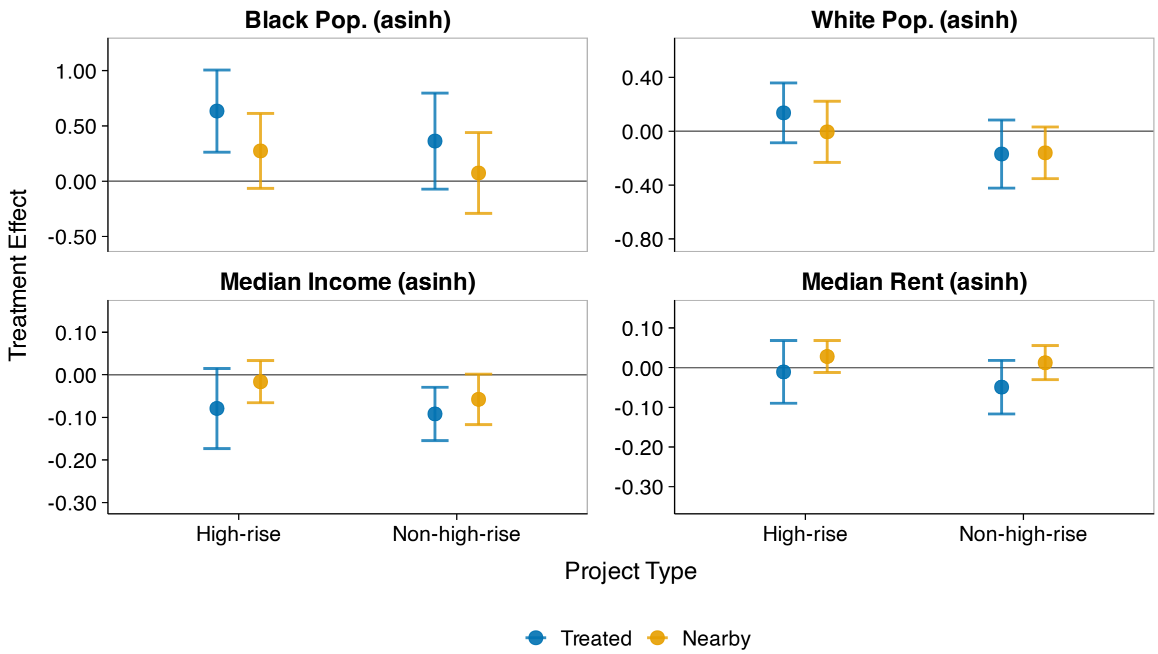
(b) Long-run Effects ($t = 3$)



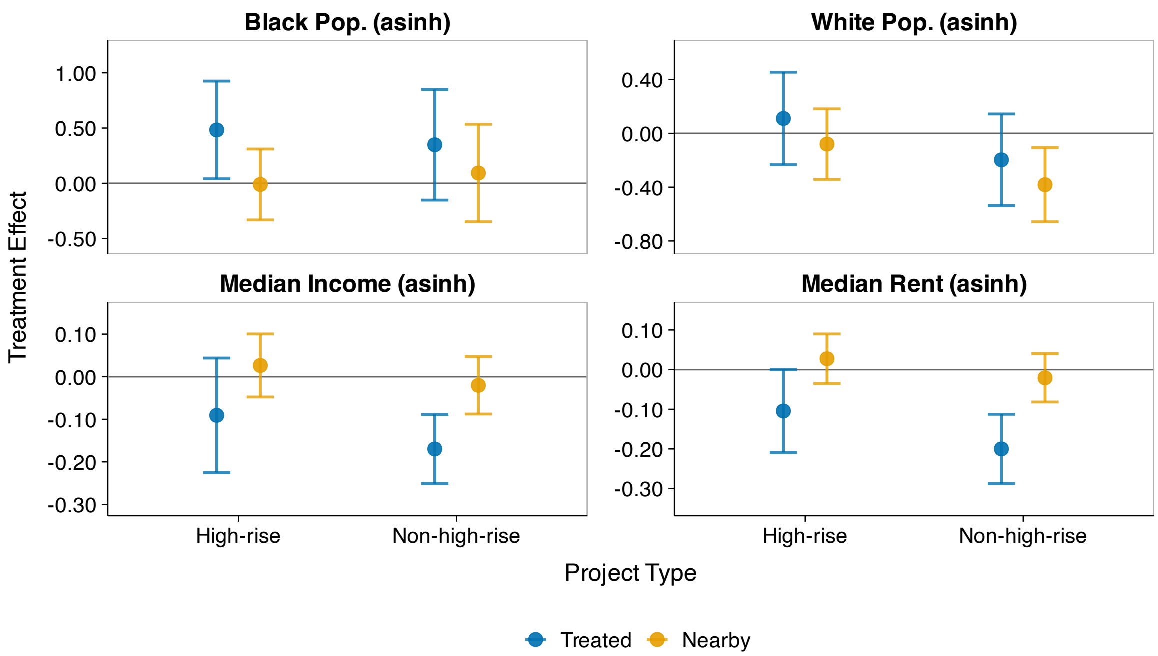
Note: This figure displays heterogeneous treatment effects by baseline Black population share, testing for neighborhood tipping dynamics. Treated neighborhoods are divided into three groups: very low (<1%, N=258), medium/tipping range (1–12%, N=186), and high (≥12%, N=360). The 12% threshold corresponds to the tipping point estimated by Card, Mas, and J. Rothstein (2008). Estimates follow Equation 3. Panel (a) shows effects at $t = 1$; panel (b) at $t = 3$. Nearby neighborhoods classified by their own baseline Black share. Population outcomes use raw counts (not asinh). Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.2 for interpretation.

Figure 17: Heterogeneity by Building Type

(a) Medium-Run Effects ($t = 1$)



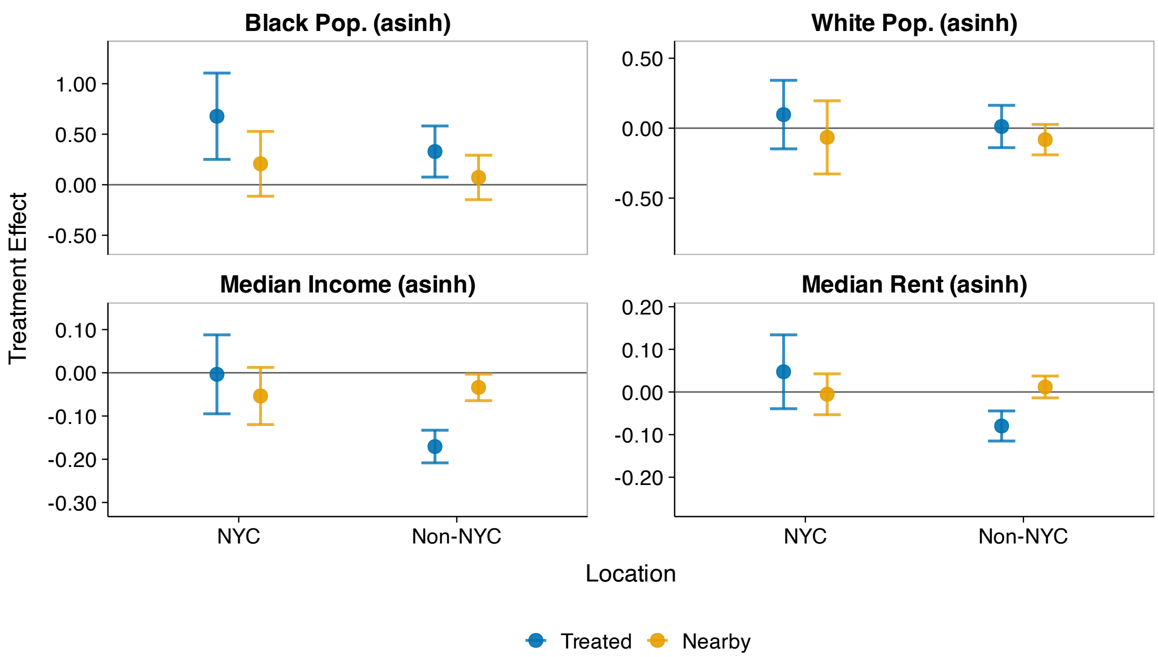
(b) Long-Run Effects ($t = 3$)



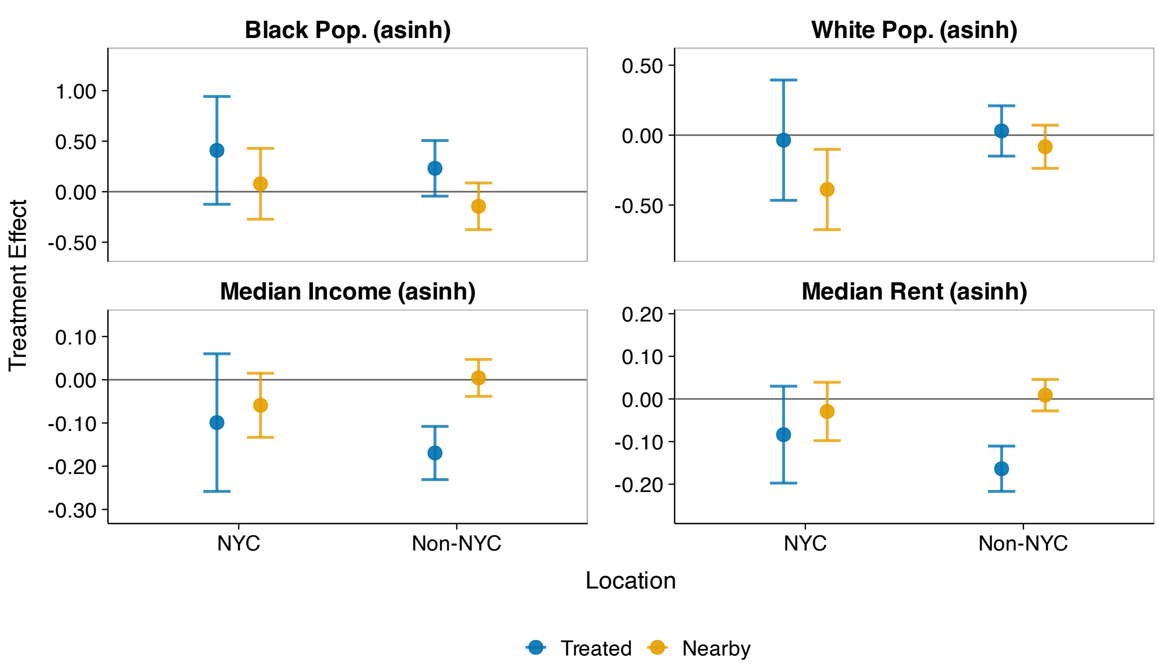
Note: This figure displays heterogeneous treatment effects by building type (high-rise vs. non-high-rise). High-rise classification based on: "elevator" designation in Chicago Housing Authority records, ≥ 7 floors in NYC Housing Authority records, or building type in HUD administrative data. Building type available for a subset of projects. Estimates follow Equation 3. Panel (a) shows effects at $t = 1$; panel (b) at $t = 3$. Population, income, and rent outcomes use asinh transformation. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.3 for interpretation and caveats regarding data coverage.

Figure 18: New York City vs Other Cities

(a) Medium-Run ($t = 1$)



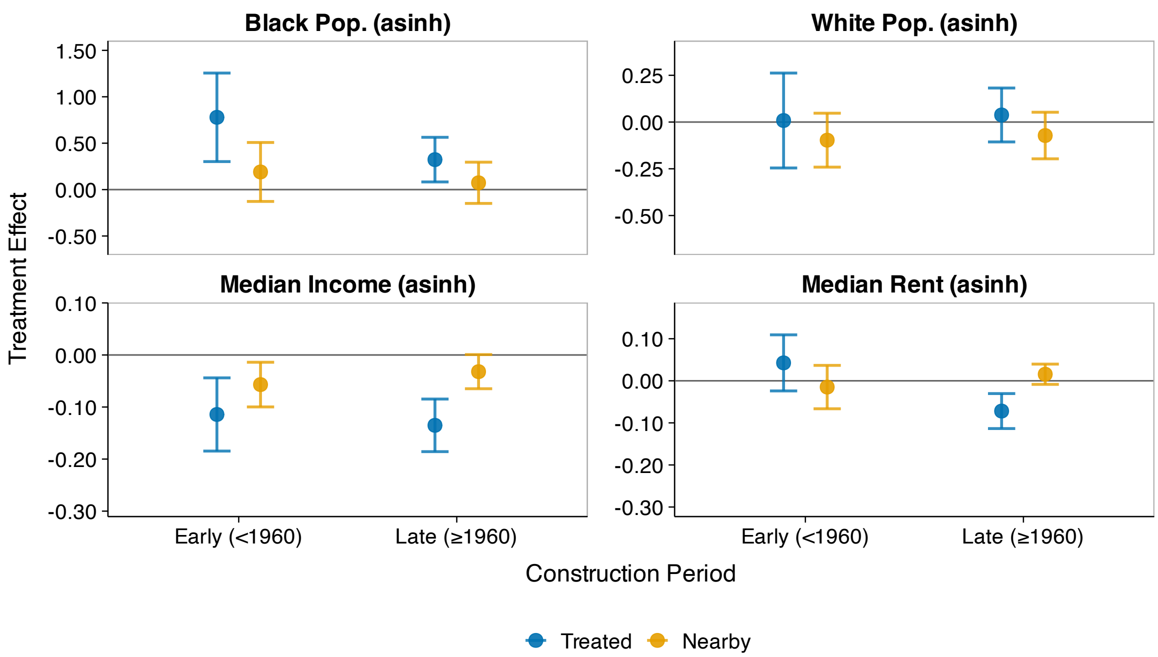
(b) Long-Run ($t = 3$)



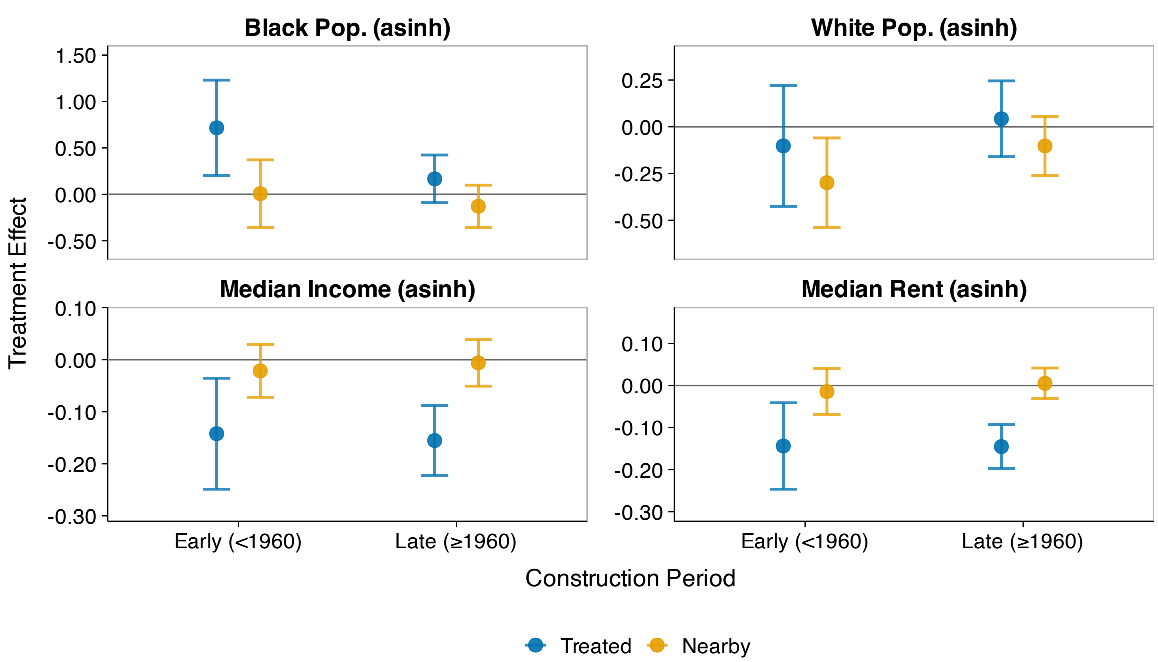
Note: This figure displays heterogeneous treatment effects comparing New York City to other cities in the sample. NYCHA was the largest U.S. housing authority and is often considered better-managed than other PHAs (Bloom 2008). Estimates follow Equation 3. Panel (a) shows effects at $t = 1$; panel (b) at $t = 3$. Population, income, and rent outcomes use asinh transformation. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.3 for interpretation.

Figure 19: Heterogeneity by Construction Decade

(a) Medium-Run ($t = 1$)



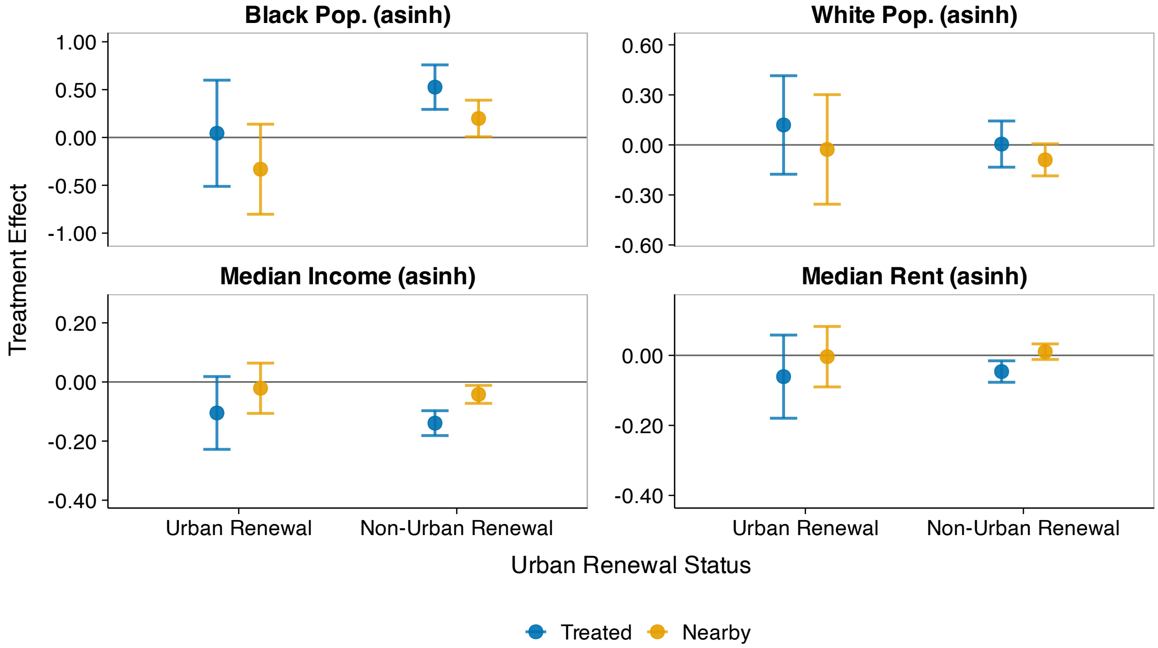
(b) Long-Run ($t = 3$)



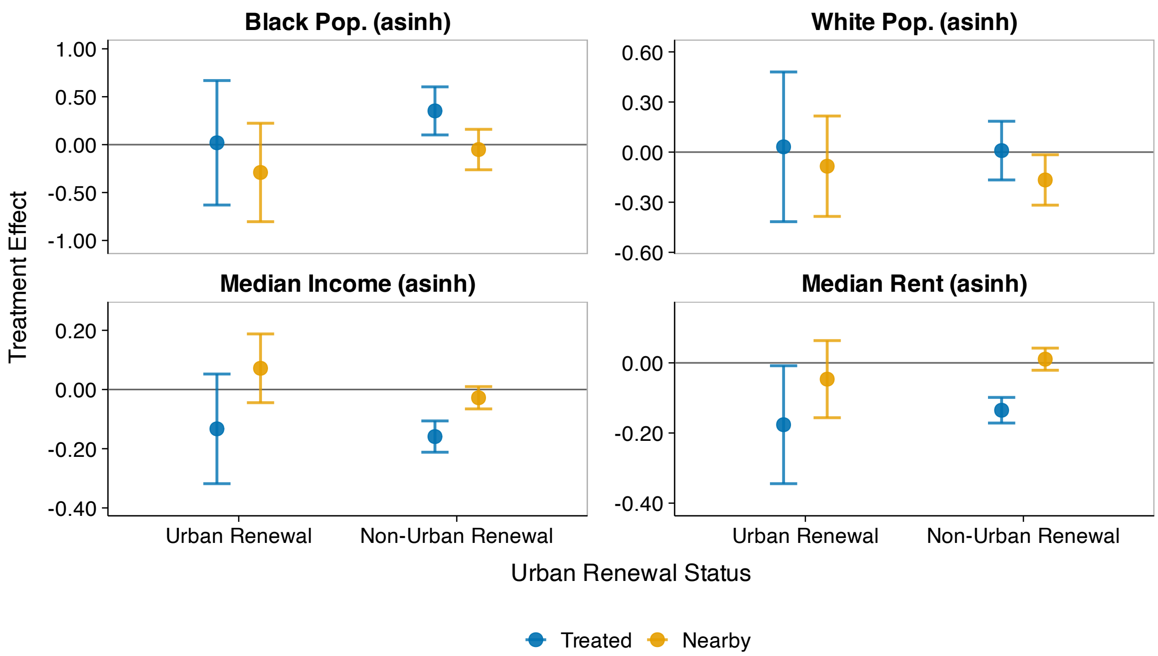
Note: This figure displays heterogeneous treatment effects by construction timing, comparing projects built before 1960 to those built 1960 or later. Early-period projects may have had different effects due to lower baseline Black population shares in urban neighborhoods during the 1940s–1950s. Estimates follow Equation 3. Panel (a) shows effects at $t = 1$; panel (b) at $t = 3$. Population, income, and rent outcomes use asinh transformation. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.4 for interpretation.

Figure 20: Heterogeneity by Urban Renewal Status

(a) Medium-Run ($t = 1$)



(b) Long-Run ($t = 3$)



Note: This figure displays heterogeneous treatment effects by urban renewal status, comparing tracts with >5% overlap with urban renewal project boundaries to non-urban renewal tracts. This analysis addresses potential confounding between the two programs and tests whether effects differ where both operated. Estimates follow Equation 3. Panel (a) shows effects at $t = 1$; panel (b) at $t = 3$. Population, income, and rent outcomes use asinh transformation. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.4 for interpretation.

Table 6: Opportunity Atlas Outcomes: Public Housing Tracts vs Matched Controls

	Mobility		Incarceration	
	(1)	(2)	(3)	(4)
Treated	-0.018***	-0.004**	0.006***	0.002**
	(0.002)	(0.002)	(0.001)	(0.001)
Num.Obs.	1540	1540	1540	1540
R2	0.758	0.877	0.660	0.736

Notes: This table reports OLS estimates of Equation 4 comparing Opportunity Atlas outcomes in public housing tracts (N=804) to matched controls. Columns (1)–(2): mean income rank in adulthood (2014–2015) for children from low-income families born 1978–1983. Columns (3)–(4): incarceration rate as of April 1, 2010. Odd columns show unconditional differences; even columns add 1980 neighborhood controls (Black share, median income, population, unemployment, rent). All specifications include matched-pair fixed effects. Standard errors clustered at the tract level in parentheses. Statistical significance: *p<0.1, **p<0.05, ***p<0.01. See Section 8 for interpretation and caveats.

Table 7: Opportunity Atlas Outcomes: Nearby Tracts vs Matched Controls

	Mobility		Incarceration	
	(1)	(2)	(3)	(4)
Nearby	-0.006***	-0.001	0.002**	0.001
	(0.002)	(0.001)	(0.001)	(0.001)
Num.Obs.	2756	2756	2756	2756
R2	0.756	0.871	0.680	0.755

Notes: This table reports OLS estimates of Equation 4 comparing Opportunity Atlas outcomes in nearby tracts (N=1,414; within 1km of public housing) to matched controls. Columns (1)–(2): mean income rank in adulthood (2014–2015) for children from low-income families born 1978–1983. Columns (3)–(4): incarceration rate as of April 1, 2010. Odd columns show unconditional differences; even columns add 1980 neighborhood controls (Black share, median income, population, unemployment, rent). All specifications include matched-pair fixed effects. Standard errors clustered at the tract level in parentheses. Statistical significance: *p<0.1, **p<0.05, ***p<0.01. See Section 8 for interpretation.

A Public Housing Data Construction

A.1 Estimating Public Housing Project Populations

This section describes the methodology for estimating the population of each public housing project using the 1977 Picture of Subsidized Households (PIC77) dataset. The PIC77 dataset contains household-level demographic information for public housing residents, including the number of households in the project, racial composition, household size, income, and age structure. However, it does not provide direct population counts.

To estimate total, white, and Black populations for each project p, I multiply the number of households for group r by the average household size:

$$\text{Population}_{pr} = \text{Households}_{pr} \times \text{Average Household Size}_{p}.$$

One challenge is that approximately 42% of projects lack household size data. To address this, I impute missing values using two categorical variables that explain substantial variation in household composition: (i) *elderly designation* (all elderly, some elderly, or none) and (ii) *racial composition* (simplified into six categories: all white, all Black, all other race, mixed, no white, and other). For each elderly designation × racial composition cell, I calculate the mean household size across all projects with non-missing data. Missing household size values are then imputed using the corresponding cell mean. This imputation strategy leverages the fact that elderly projects have systematically smaller household sizes, and household size varies by the racial composition of the project.

This approach assumes that average household size is constant across racial groups within each project. While household size may vary by race in the general population, this assumption is reasonable within public housing projects where unit size allocations are determined by family size rather than race, and eligibility criteria are applied uniformly across racial groups.

To validate the imputation strategy, I conducted a hold-out test where 20% of projects with observed household size were randomly set to missing and then imputed using the cell means calculated from the remaining 80%. The imputation has an R^2 of 0.46 and is essentially unbiased, with a mean error of -0.01 persons.

B Census Tract Data Construction and Harmonization

B.1 Data Sources and Variable Availability

I construct tract-level neighborhood characteristics from two primary sources: published Census tabulations from NHGIS (Manson et al. 2022) and full-count Census microdata from IPUMS (Ruggles et al. 2022). The two sources play complementary roles.

For 1950 onward, NHGIS provides comprehensive tract-level tabulations for all tracted areas, including population by race, housing characteristics, and socioeconomic measures. I use these tabulations as the primary data source for these decades.

For 1930 and 1940, NHGIS tract coverage is limited to select cities. I therefore supplement NHGIS data with tract-level aggregates constructed from full-count Census microdata, using enumeration district to tract crosswalks from the Urban Transition Historical GIS Project (J. R. Logan et al. 2024). For tracts where both sources are available, I use the NHGIS tabulations; for tracts covered only in the full-count data, I use the microdata-based estimates. This approach maximizes geographic coverage while maintaining consistency with published Census figures where available.

Table 8 summarizes the data format for key variables by decade. The format determines how I calculate tract-level values when harmonizing to 1950 boundaries (see Section B.2).

Table 8: Census Tract Variable Data Formats by Decade

Variable	1930	1940	1950	1960	1970	1980	1990	2000	2010
Population by race	Count[a]	Count[a]	Count	Count	Count	Count	Count	Count	Count
Median income	FC	FC	Dist.	Dist.	Dist.	Rep.	Rep.	Rep.	Rep.
Median rent	Dist.[a]	Rep.	Rep.	Dist.	Dist.	Rep.	Rep.	Rep.	Rep.
Median home value	Dist.[a]	Rep.	Rep.	Dist.	Dist.	Rep.	Rep.	Rep.	Rep.
Employment/LFP	FC	FC	Count	Count	Count	Count	Count	Count	Count
Education	—	FC	Count	Count	Count	Count	Count	Count	Count

Notes: "Count" = raw counts from NHGIS (Manson et al. 2022). "Rep." = reported median from NHGIS. "Dist." = median calculated from frequency distributions. "FC" = constructed from full-count Census microdata (Ruggles et al. 2022). [a]NHGIS coverage limited to select cities; full-count microdata used for remaining cities.

The Census Bureau's reporting practices for median variables changed substantially over time. Median family income was first reported at the tract level in 1980; earlier decades provide only frequency distributions (1950–1970) or require construction from microdata (1930–1940). For housing variables, the Census reported medians in 1940, 1950, and 1980 onward, but provided only distributions in 1960 and 1970.

For variables reported as frequency distributions, I calculate medians using midpoint interpolation (described below). For all economic variables, I construct a composite measure that uses reported medians when available and calculates medians from distributions otherwise, ensuring consistent coverage across decades.

B.1.1 Calculating Medians from Categorical Distributions

For census years that report only frequency distributions across income, rent, or home value ranges, I calculate medians using a midpoint interpolation method. For each tract, I first calculate cumulative household counts across the income, rent, or value ranges reported by the Census Bureau. I then identify the first range where the cumulative frequency exceeds 50% of total households. Finally, I assign the midpoint of that range as the median value.

Formally, let n_i denote the number of households in range i, with bounds $[L_i, U_i]$. The median is calculated as:

$$\text{Median} = \frac{L_m + U_m}{2} \tag{5}$$

where m is the first range satisfying $\sum_{i=1}^{m} n_i \geq \frac{1}{2} \sum_{i=1}^{N} n_i$. For top-coded categories (e.g., "$25,000 or more"), I use the lower bound since the upper bound is undefined.

For example, the 1960 Census reports median income as 13 frequency bins: <$1,000, $1,000-$2,000, $2,000-$3,000, ..., $25,000+. If a tract has 1,000 families and the cumulative count first exceeds 500 at the $5,000-$7,500 range, I assign median income as $6,250. Similarly, for 1970 median rent, the Census provides 15 bins ranging from <$20 to $500+, and the same midpoint interpolation method is applied.

When crosswalking census tracts across decades to consistent 1950 boundaries (see Section B.2 for details), the methodology differs depending on whether medians are reported or calculated. For years with reported medians (income: 1980+; rent/value: 1940, 1950, 1980+), I use population-weighted averaging of median values when aggregating across source tracts. For years with distributions (income: 1950-1970; rent/value: 1930, 1960, 1970), I reallocate the count of households in each income or rent range using area weights, sum across source tracts, and then calculate the median from the resulting distribution. This distinction is critical because medians are intensive variables that should be averaged rather than summed, as explained in Section B.2.

All monetary values are converted to real 2000 dollars using the Consumer Price Index (CPI-U) from Officer and Williamson (2025). The adjustment is applied after calculating medians (for distribution years) or using reported medians (for other years), but before crosswalking to 1950 tract boundaries.

B.2 Tract Boundary Harmonization

This section describes the geographic harmonization of Census tract-level data to consistent boundaries. My baseline analysis harmonizes all tract-level data from 1930 to 2010 to 1950 Census tract boundaries using an area reweighting approach following Eckert et al. (2020).

I proceed as follows. First, for each source year, I spatially intersect census tract shapefiles with 1950 tract boundaries to identify overlapping areas. Next, for each source tract i, I calculate weights as the proportion of the tract's area that falls within each 1950 tract j:

$$w_{ij} = \frac{\text{Area}(\text{Tract}_i \cap \text{Tract}_{1950,j})}{\text{Area}(\text{Tract}_i)}$$

Weights are normalized to sum to one for each source tract to account for minor geometric inconsistencies.

Finally, I use these weights to estimate count and median variables in the 1950 tract geography. This proceeds slightly differently for count-based variables and median-based variables.

For each 1950 tract j, count variables such as population, housing units, and employment are harmonized by simply summing the weighted contributions from all overlapping source tracts:

$$X_j = \sum_{i \in I_j} w_{ij} \cdot X_i$$

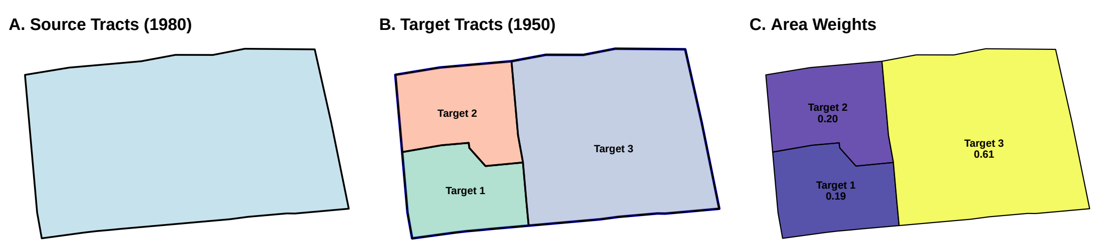
where X_i represents the count variable in source tract i and I_j is the set of all source tracts overlapping with 1950 target tract j. This method assumes that outcome variables are uniformly distributed within each source tract, which may introduce measurement error in areas where populations are spatially concentrated within tract boundaries. For instance, if a source tract contains a spatially concentrated Black population but overlaps evenly with multiple 1950 tracts, the method would allocate residents uniformly across those tracts, potentially attenuating spatial estimates of segregation.

For median variables such as median income, median rent, and median home value, tract-level estimates are calculated as weighted averages across source tracts:

$$\tilde{X}_j = \frac{\sum_{i \in I_j} w_{ij} \cdot \tilde{X}_i}{\sum_{i \in I_j} w_{ij}}$$

The weights w_{ij} are based solely on area overlap. This follows from the uniform-within-tract assumption described above: under the assumption that population (and the relevant sub-populations for each median) are uniformly distributed within each source tract, the share of a tract's population falling into a target tract is proportional to the area share w_{ij}.[11] This weighted-average approach is mathematically equivalent to using target-tract area shares, as recommended by Eckert et al. (2020) for intensive variables, because the denominator renormalizes contributions to sum to one for each target tract. This approach is applied to median family income (1980, 1990, 2000), median contract rent (1940, 1980, 1990, 2000), and median home value (1940, 1980, 1990, 2000).

(a) Split Pattern: One 1980 tract → Multiple 1950 tracts



(b) Merge Pattern: Multiple 1980 tracts → One 1950 tract

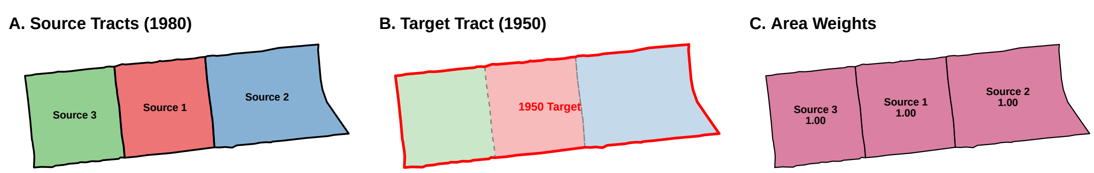


Figure 21: Census Tract Harmonization Examples

Note: These figures demonstrate the area-reweighting methodology for harmonizing 1980 census tracts to 1950 boundaries in Chicago. Panel (a) shows a split pattern where one 1980 source tract is divided across multiple 1950 target tracts, with weights representing the proportion of area allocated to each target. Panel (b) shows a merge pattern where multiple 1980 source tracts contribute to a single 1950 target tract.

[11] Population weighting would require sub-tract spatial information on these subpopulations (renter households for median rent, families for median income), which is unavailable.

C Robustness Checks: Alternative Matching Specifications and Standard Errors

This appendix presents full event study coefficient tables for alternative matching specifications discussed in the main text.

C.1 Alternative Matching Specifications

Here, I present the full set of coefficients from two alternative specifications that address potential limitations of the baseline matching approach. First, I present results using the baseline matching specification but adding a caliper restriction of 0.2 standard deviations of the propensity score to drop poor matches to improve covariate balance. Second, I run a specification in which I match public housing and nearby tracts to control tracts from different metropolitan areas, rather than within the same county. This specification expands the donor pool for each tract, potentially allowing for closer matches on pre-treatment characteristics, particularly for treated tracts in counties with limited control tracts.

Table 9: Caliper Matching: Treated Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -20	0.018 (0.113)	-0.015 (0.019)	-0.063 (0.051)	0.006 (0.014)
t = +0	0.580*** (0.112)	0.145*** (0.021)	0.086* (0.047)	0.033*** (0.010)
t = +10	0.589*** (0.133)	0.161*** (0.027)	0.060 (0.070)	0.057*** (0.016)
t = +20	0.513*** (0.141)	0.158*** (0.031)	0.038 (0.084)	0.063*** (0.019)
t = +30	0.386*** (0.138)	0.159*** (0.033)	0.032 (0.087)	0.061*** (0.019)

Notes: Event study coefficients from Equation 3 for treated neighborhoods using caliper matching (baseline specification plus 0.2 SD propensity score caliper). Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 10: Caliper Matching: Treated Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -20	0.003 (0.016)	-0.034* (0.018)	0.005 (0.005)	-0.005 (0.003)
t = +0	-0.076*** (0.014)	-0.025** (0.012)	-0.002 (0.004)	0.008*** (0.003)
t = +10	-0.140*** (0.020)	-0.063*** (0.017)	-0.011* (0.007)	0.004 (0.004)
t = +20	-0.163*** (0.022)	-0.099*** (0.019)	-0.023*** (0.008)	0.011** (0.005)
t = +30	-0.175*** (0.028)	-0.163*** (0.023)	-0.026*** (0.009)	0.014*** (0.005)

Notes: Event study coefficients for treated neighborhoods using caliper matching with propensity score caliper restriction. Outcomes include asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Asinh denotes the inverse hyperbolic sine transformation. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01.

Table 11: Caliper Matching: Spillover Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -20	-0.106 (0.090)	0.008 (0.013)	-0.073* (0.039)	0.003 (0.008)
t = +0	0.223*** (0.085)	0.011 (0.011)	0.027 (0.042)	0.025*** (0.008)
t = +10	0.183* (0.100)	-0.013 (0.017)	-0.064 (0.054)	0.040*** (0.012)
t = +20	0.049 (0.105)	-0.021 (0.018)	-0.119* (0.069)	0.047*** (0.015)
t = +30	-0.055 (0.104)	-0.027 (0.021)	-0.142* (0.079)	0.050*** (0.016)

Notes: Event study coefficients for spillover neighborhoods (adjacent to treated tracts) using caliper matching with propensity score caliper restriction. Outcomes include asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Asinh denotes the inverse hyperbolic sine transformation. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01. Significance stars are assigned using full-precision coefficients and standard errors; displayed values are rounded to three decimal places.

Table 12: Caliper Matching: Spillover Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -20	0.013 (0.009)	0.002 (0.012)	0.003 (0.004)	-0.002 (0.002)
t = +0	-0.025** (0.010)	0.008 (0.009)	-0.003 (0.003)	-0.003 (0.003)
t = +10	-0.029* (0.015)	0.004 (0.011)	-0.000 (0.006)	-0.001 (0.003)
t = +20	-0.022 (0.018)	-0.008 (0.014)	0.000 (0.007)	-0.002 (0.004)
t = +30	-0.011 (0.021)	0.001 (0.017)	0.002 (0.008)	0.004 (0.004)

Notes: Event study coefficients for spillover neighborhoods (adjacent to treated tracts) using caliper matching with propensity score caliper restriction. Outcomes include asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Asinh denotes the inverse hyperbolic sine transformation. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01.

Table 13: No CBSA Restriction: Treated Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -20	0.082 (0.080)	-0.028* (0.016)	-0.034 (0.039)	0.001 (0.009)
t = +0	0.407*** (0.090)	0.121*** (0.018)	0.087** (0.043)	0.022** (0.009)
t = +10	0.554*** (0.108)	0.155*** (0.025)	0.074 (0.068)	0.047*** (0.014)
t = +20	0.506*** (0.119)	0.169*** (0.031)	0.009 (0.075)	0.064*** (0.016)
t = +30	0.417*** (0.126)	0.164*** (0.036)	0.004 (0.081)	0.070*** (0.017)

Notes: Event study coefficients for treated neighborhoods using propensity score matching that restricts controls to different metropolitan areas. Outcomes include asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01.

Table 14: No CBSA Restriction: Treated Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -20	0.007 (0.013)	0.032* (0.017)	-0.001 (0.005)	-0.001 (0.006)
t = +0	-0.070*** (0.013)	-0.010 (0.012)	-0.000 (0.004)	0.001 (0.006)
t = +10	-0.122*** (0.019)	-0.045** (0.018)	-0.003 (0.006)	-0.004 (0.007)
t = +20	-0.122*** (0.023)	-0.080*** (0.021)	-0.009 (0.009)	-0.001 (0.007)
t = +30	-0.093*** (0.028)	-0.119*** (0.023)	-0.016 (0.010)	0.007 (0.006)

Notes: Event study coefficients for treated neighborhoods using propensity score matching that restricts controls to different metropolitan areas. Outcomes include asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01.

Table 15: No CBSA Restriction: Spillover Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -20	-0.129* (0.074)	-0.023* (0.013)	-0.027 (0.029)	-0.005 (0.007)
t = +0	0.051 (0.074)	-0.006 (0.011)	-0.022 (0.031)	0.010 (0.007)
t = +10	0.106 (0.102)	-0.016 (0.017)	-0.054 (0.052)	0.021* (0.011)
t = +20	-0.018 (0.109)	-0.016 (0.024)	-0.080 (0.068)	0.029** (0.013)
t = +30	-0.098 (0.106)	-0.009 (0.027)	-0.098 (0.073)	0.037** (0.014)

Notes: Event study coefficients for spillover neighborhoods (adjacent to treated tracts) using propensity score matching that restricts controls to different metropolitan areas. Outcomes include asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01.

Table 16: No CBSA Restriction: Spillover Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -20	0.006 (0.010)	0.011 (0.015)	-0.004 (0.004)	-0.004 (0.005)
t = +0	-0.048*** (0.009)	-0.023** (0.010)	-0.000 (0.003)	-0.004 (0.006)
t = +10	-0.050*** (0.013)	-0.017 (0.013)	0.004 (0.005)	-0.007 (0.005)
t = +20	-0.042** (0.017)	-0.014 (0.017)	0.008 (0.007)	-0.008 (0.005)
t = +30	-0.009 (0.019)	-0.010 (0.019)	0.009 (0.009)	-0.005 (0.006)

Notes: Event study coefficients for spillover neighborhoods (adjacent to treated tracts) using propensity score matching that restricts controls to different metropolitan areas. Outcomes include asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01.

C.2 Low-Intensity Metro Controls: Addressing SUTVA Concerns

This section presents results using control neighborhoods drawn only from metropolitan areas with low public housing intensity, as described in Section 6.4. I calculate each metro's intensity as total public housing units divided by the 1950 population of sampled neighborhoods, expressed per 1,000 residents. I restrict the control pool to metros in the bottom quintile of intensity (fewer than 3.7 units per 1,000). Treated neighborhoods from all metros are matched to controls in these low-intensity metros, with matches required to come from different metropolitan areas.

Table 17: Low-Intensity Controls: Treated Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -20	0.105 (0.126)	-0.050* (0.026)	-0.053 (0.042)	0.001 (0.014)
t = +0	0.344*** (0.100)	0.136*** (0.028)	0.060 (0.053)	0.007 (0.013)
t = +10	0.473*** (0.150)	0.099* (0.054)	-0.043 (0.095)	0.041** (0.020)
t = +20	0.599*** (0.186)	0.105 (0.071)	-0.159 (0.128)	0.077*** (0.027)
t = +30	0.566*** (0.207)	0.122 (0.086)	-0.121 (0.152)	0.083*** (0.031)

Notes: Event study coefficients for treated neighborhoods using propensity score matching that restricts controls to metropolitan areas in the bottom quintile of public housing intensity (<3.7 units per 1,000 residents). Matches are required to be from different metropolitan areas. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 18: Low-Intensity Controls: Treated Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -20	0.017 (0.016)	0.184 (0.136)	-0.004 (0.007)	-0.002 (0.007)
t = +0	-0.038 (0.029)	-0.085* (0.045)	0.009* (0.005)	-0.012 (0.008)
t = +10	-0.060*** (0.023)	-0.097** (0.043)	0.026** (0.012)	-0.019** (0.009)
t = +20	-0.043 (0.032)	-0.120** (0.051)	0.041** (0.017)	-0.016* (0.009)
t = +30	0.054 (0.038)	-0.121*** (0.045)	0.057*** (0.020)	-0.009 (0.011)

Notes: Event study coefficients for treated neighborhoods using propensity score matching that restricts controls to metropolitan areas in the bottom quintile of public housing intensity. Outcomes include asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 19: Low-Intensity Controls: Spillover Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -20	-0.052 (0.095)	-0.036** (0.016)	-0.052 (0.047)	-0.010 (0.010)
t = +0	-0.024 (0.086)	0.003 (0.019)	0.036 (0.058)	0.005 (0.010)
t = +10	-0.017 (0.132)	0.006 (0.032)	-0.066 (0.076)	0.017 (0.016)
t = +20	-0.142 (0.158)	-0.018 (0.042)	-0.195* (0.111)	0.037* (0.022)
t = +30	-0.141 (0.180)	0.007 (0.057)	-0.164 (0.137)	0.042 (0.027)

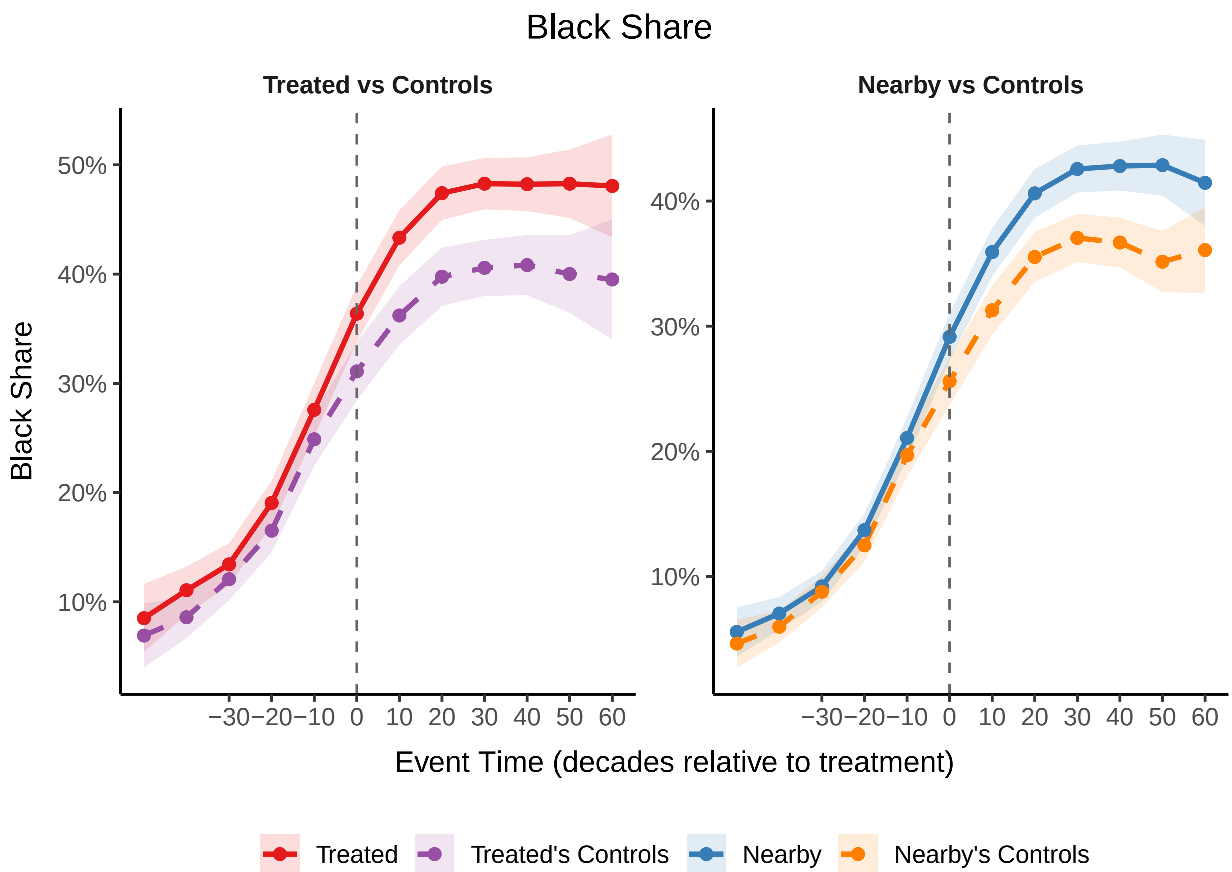
Notes: Event study coefficients for spillover neighborhoods (adjacent to treated tracts) using propensity score matching that restricts controls to metropolitan areas in the bottom quintile of public housing intensity. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 20: Low-Intensity Controls: Spillover Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -20	0.002 (0.011)	-0.065 (0.041)	-0.013** (0.005)	-0.005 (0.006)
t = +0	-0.016 (0.010)	-0.046 (0.043)	0.013*** (0.005)	-0.020*** (0.006)
t = +10	0.025 (0.015)	0.002 (0.049)	0.033*** (0.008)	-0.032*** (0.007)
t = +20	0.060*** (0.022)	0.021 (0.053)	0.052*** (0.012)	-0.032*** (0.006)
t = +30	0.158*** (0.026)	0.054 (0.053)	0.073*** (0.015)	-0.027*** (0.007)

Notes: Event study coefficients for spillover neighborhoods using propensity score matching that restricts controls to metropolitan areas in the bottom quintile of public housing intensity. Outcomes include asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Standard errors account for spatial correlation following Conley (1999) with 2km cutoff. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Figure 22: Black Population Share: Treated and Nearby Neighborhoods vs Matched Controls Over Event Time



Note: Mean Black population share for treated neighborhoods and their matched controls (left panel) and nearby neighborhoods and their matched controls (right panel), aligned by event time (decades relative to public housing construction). Shaded bands show 95% confidence intervals. Both treated and control neighborhoods show increasing Black shares over time, consistent with the Great Migration. The treatment effect is identified by the differential increase in treated relative to controls.

C.3 Baseline Specification: Tract-Level Clustered Standard Errors

This section presents event study coefficients for the baseline matching specification using standard errors clustered at the census tract level, rather than using Conley (1999) spatially-correlated standard errors, as used in the main text.

Table 21: Baseline Matching: Treated Neighborhoods, Population and Demographics (Tract-Clustered SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -20	0.040 (0.080)	0.000 (0.016)	-0.007 (0.038)	-0.002 (0.010)
t = +0	0.416*** (0.080)	0.125*** (0.017)	0.095** (0.038)	0.026*** (0.008)
t = +10	0.411*** (0.097)	0.133*** (0.023)	0.032 (0.052)	0.044*** (0.011)
t = +20	0.367*** (0.101)	0.141*** (0.026)	0.028 (0.064)	0.050*** (0.013)
t = +30	0.273*** (0.102)	0.147*** (0.027)	0.014 (0.067)	0.050*** (0.014)

Notes: Event study coefficients from Equation 3 for treated neighborhoods (N=804) using baseline matching (exact on county and urban renewal). Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Standard errors clustered at the census tract level in parentheses (cf. Conley SEs in main text). Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 22: Baseline Matching: Treated Neighborhoods, Economic and Housing Outcomes (Tract-Clustered SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -20	0.005 (0.012)	-0.024* (0.013)	0.006 (0.004)	-0.002 (0.003)
t = +0	-0.077*** (0.012)	-0.004 (0.010)	-0.003 (0.004)	0.003 (0.003)
t = +10	-0.131*** (0.016)	-0.050*** (0.014)	-0.010* (0.005)	0.002 (0.003)
t = +20	-0.152*** (0.019)	-0.085*** (0.016)	-0.021*** (0.006)	0.008* (0.004)
t = +30	-0.153*** (0.022)	-0.145*** (0.019)	-0.025*** (0.007)	0.010** (0.004)

Notes: Event study coefficients for treated neighborhoods using baseline propensity score matching with exact matching on CBSA and urban renewal status. Outcomes include asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Standard errors are clustered at the census tract level. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01.

Table 23: Baseline Matching: Spillover Neighborhoods, Population and Demographics (Tract-Clustered SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -20	-0.055 (0.066)	0.004 (0.012)	-0.034 (0.027)	-0.002 (0.006)
t = +0	0.157*** (0.060)	-0.002 (0.010)	0.007 (0.027)	0.022*** (0.006)
t = +10	0.102 (0.073)	-0.023 (0.015)	-0.078** (0.038)	0.033*** (0.009)
t = +20	-0.013 (0.079)	-0.026 (0.016)	-0.127*** (0.045)	0.037*** (0.010)
t = +30	-0.094 (0.078)	-0.036** (0.018)	-0.152*** (0.050)	0.041*** (0.011)

Notes: Event study coefficients for spillover neighborhoods (adjacent to treated tracts) using baseline propensity score matching with exact matching on CBSA and urban renewal status. Outcomes include asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Standard errors are clustered at the census tract level. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01.

Table 24: Baseline Matching: Spillover Neighborhoods, Economic and Housing Outcomes (Tract-Clustered SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -20	0.007 (0.008)	0.005 (0.011)	0.004 (0.003)	-0.002 (0.002)
t = +0	-0.030*** (0.009)	0.011 (0.008)	-0.002 (0.003)	-0.003* (0.002)
t = +10	-0.038*** (0.012)	0.008 (0.009)	-0.001 (0.004)	-0.003 (0.002)
t = +20	-0.023* (0.013)	0.000 (0.011)	-0.000 (0.004)	-0.004 (0.003)
t = +30	-0.010 (0.014)	0.000 (0.012)	-0.000 (0.005)	0.004 (0.003)

Notes: Event study coefficients for spillover neighborhoods (adjacent to treated tracts) using baseline propensity score matching with exact matching on CBSA and urban renewal status. Outcomes include asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Standard errors are clustered at the census tract level. Reference period is one decade before public housing construction ($t = -1$). Statistical significance is denoted by: *p<0.1, **p<0.05, ***p<0.01.

D Chicago High-Resolution Land Value Analysis

This appendix presents a supplementary spatial analysis of the effects of public housing on land values in Chicago using a high-resolution panel of land values. I use a dataset from Ahlfeldt and McMillen (2018) that provides nearly-decadal data on land values at a 300×300 foot grid-cell level from 1913 to 2010, sourced from *Olcott's Land Values - Blue Book of Chicago*. While my baseline analysis estimates the effects of public housing on Census self-reported rents, assessed land values may provide additional insights into the economic impacts of these projects. Furthermore, the spatial granularity of these data is particularly well-suited for a spatial difference-in-differences design using concentric rings around project locations.

D.1 Empirical Strategy

I estimate the effects of public housing on land values using a stacked spatial difference-in-differences design following Blanco and Neri (2025), leveraging the high-resolution grid-cell data to analyze how public housing projects affected land values in concentric rings around project locations. As discussed in Section 5, the locations of public housing projects were determined by a combination of neighborhood characteristics. Here, given the spatial granularity of the data, I define the comparison points based on proximity to public housing project locations.

For each project, I compare the evolution of land values within a ring of a given radius around each project to those in a control ring farther away. I define treatment rings as 200m concentric buffers around each public housing project, and define the control rings as the ring of grid points 800-1000m away from the project. The data and estimation are organized in a stacked difference-in-differences framework with each public housing project treated as a separate "sub-experiment" (Wing, Freedman, and Hollingsworth 2024). The identifying assumption is that the trends in land values would have been similar between the treatment rings (0-800m) and the control rings (800-1000m) around each project in the absence of public housing construction.

I estimate the following event study at the grid cell i, public housing project g, and year t level:

$$\text{llv}_{igt} = \sum_{k\neq -1} \sum_{r\in R} \beta_{kr} \mathbf{1}[\text{event\_time}_{gt} = k] \times \mathbf{1}[\text{Ring}_{ig} = r] + \alpha_{gt} + \mu_{gr} + \gamma_{igt} + \epsilon_{igt} \tag{6}$$

where i indexes grid cells, g indexes public housing projects, t indexes years, k indexes event time, and r indexes spatial rings. The variable llv_{igt} is the log land value in grid cell i at time t around public housing project g, event_time_{gt} is the event time relative to project g's opening, and Ring_{ig} indicates the spatial ring type (e.g., 0-200m, 200-400m, etc.). The coefficients β_{kr} capture the effects of public housing projects on land values at different event times k and across different spatial rings r. The term α_{gt} represents project-by-year fixed effects that ensure identification comes from comparing grid cells within each project over time, rather than across different projects. The term μ_{gr} represents project-by-ring fixed effects that control for baseline differences across spatial rings within each project. The term γ_{igt} includes additional controls—urban renewal status and proximity to an interstate highway—interacted with project identifiers to allow effects to vary flexibly within each project.

D.2 Results

Figure 23 presents the main results from the spatial land value analysis. The baseline specification shows that public housing projects, on average, had positive effects on local land values, particularly in the immediate vicinity of the projects. These effects mostly do not reach statistical significance, however.

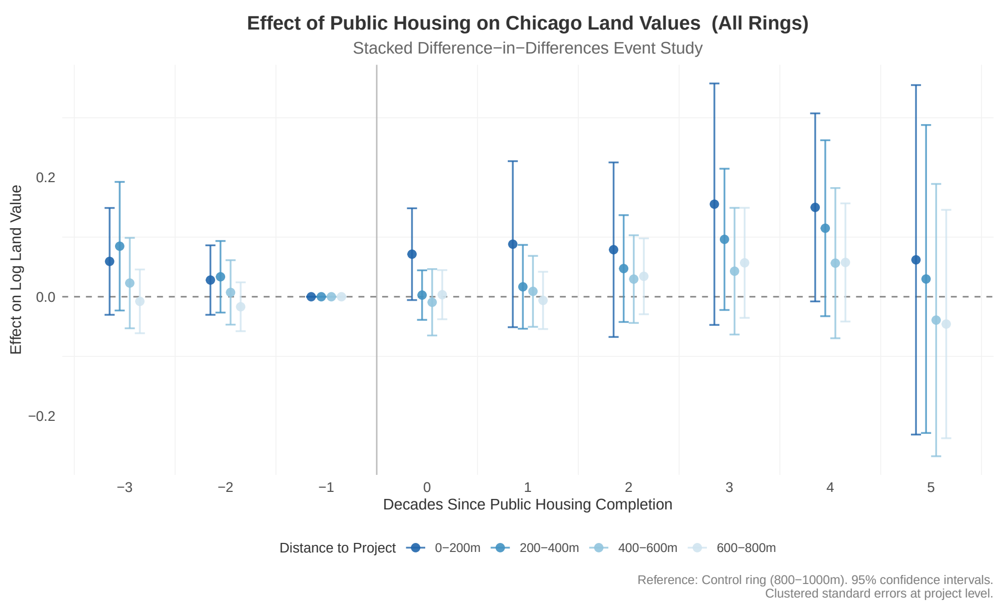


Figure 23: Effect of Public Housing on Chicago Land Values - Main Event Study

Note: Event study estimates of public housing effects on log land values in Chicago (N=73 projects), using a stacked spatial difference-in-differences design. Estimates shown for 200-meter concentric rings around projects (0–200m, 200–400m, 400–600m, 600–800m), with grid cells 800–1000m from projects serving as the control ring. The reference period is one period before construction (event time $= -1$). Vertical bars represent 95% confidence intervals with standard errors clustered at the project level. Land value data from Ahlfeldt and McMillen (2018), measured at 300×300 foot grid cells from 1913–2010.

Figure 24 examines how these effects vary based on the size of the projects. I estimate the main specification separately for projects above the median project size (202 units) and those at or below it, comparing 55 smaller projects to 18 larger developments. This analysis reveals substantial heterogeneity in the impacts on neighborhood land values. In particular, I find some evidence that smaller projects seemed to have more persistent positive effects on local land values.

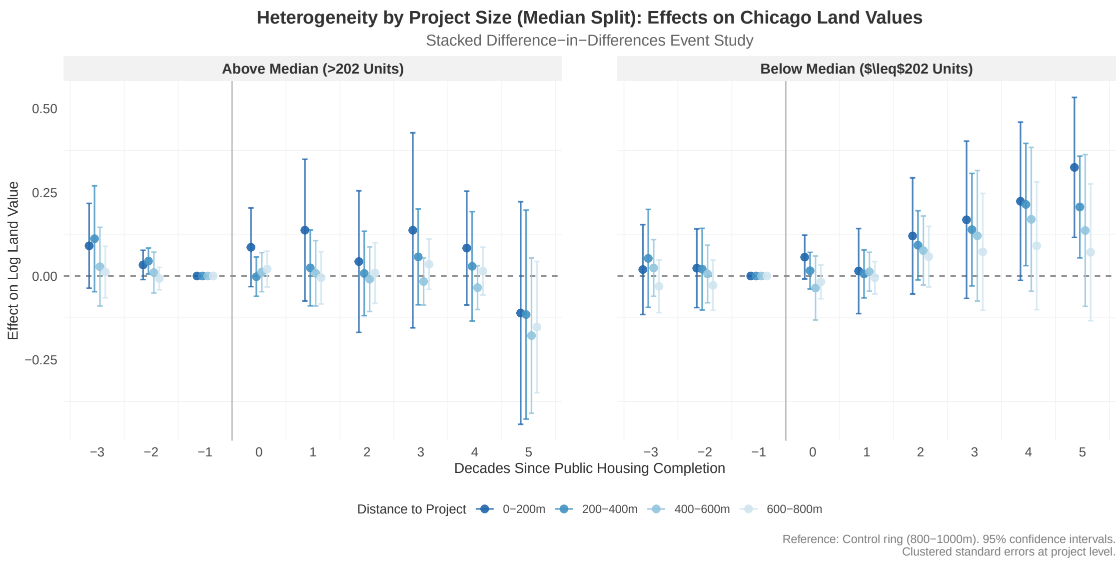


Figure 24: Heterogeneity by Project Size - Median Split Event Study

Note: Event study estimates comparing the effects of smaller (≤202 units, N=55 projects) and larger (>202 units, N=18 projects) public housing projects on log land values in Chicago. Estimates are shown separately for each project size category across the same concentric rings as in Figure 23. Vertical bars represent 95% confidence intervals with standard errors clustered at the project level. Source: Land value data from Ahlfeldt and McMillen (2018).